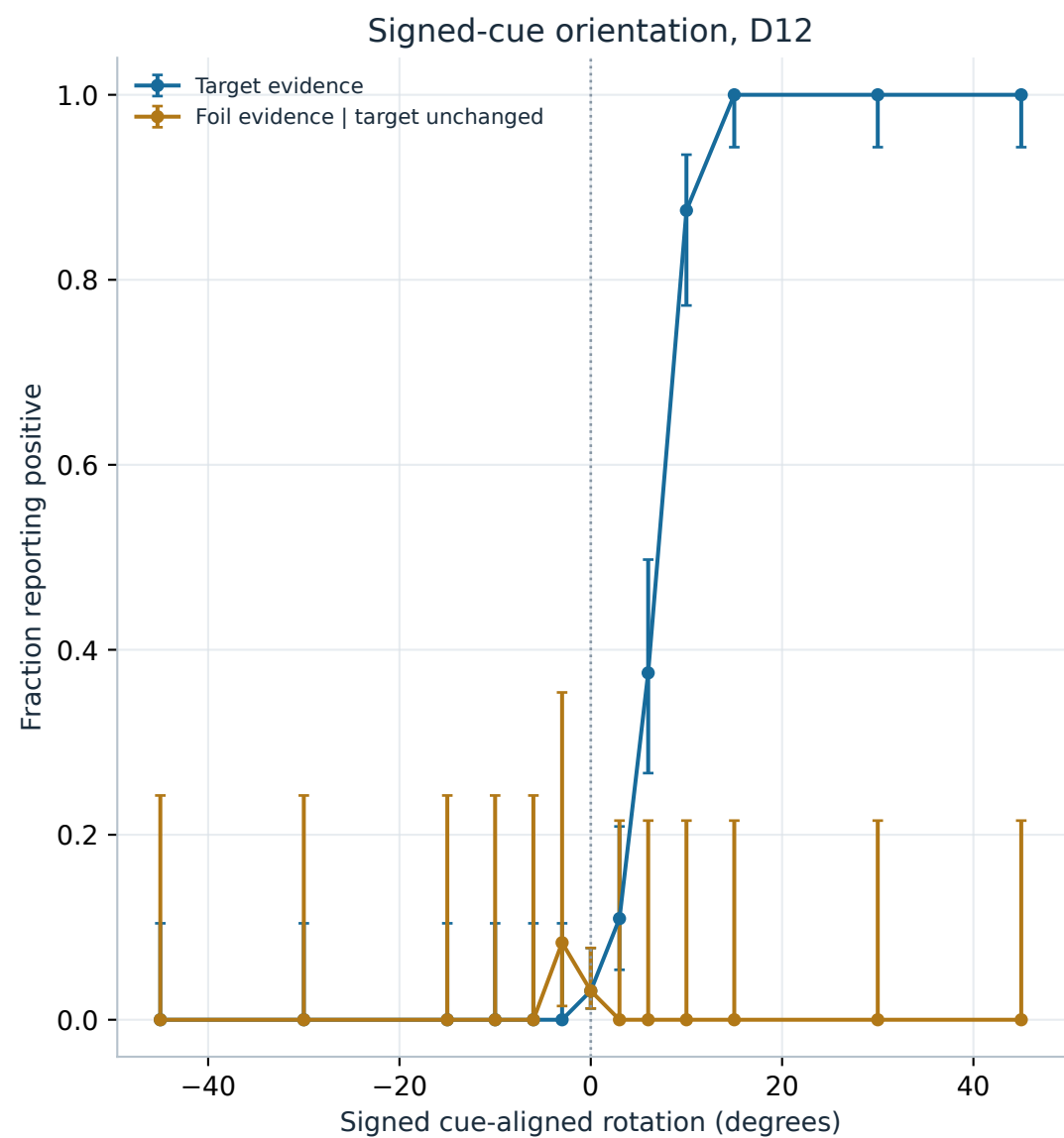
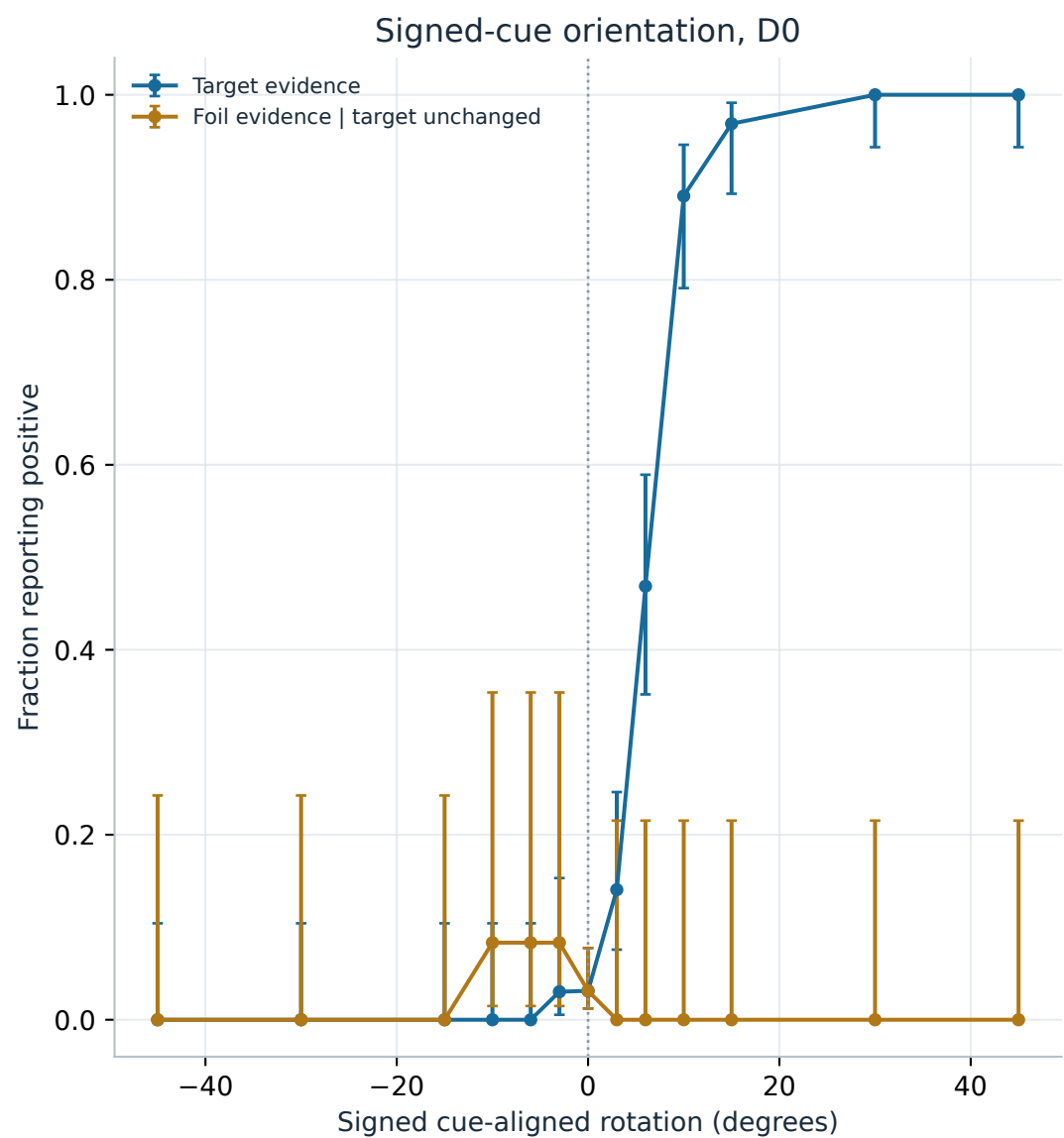
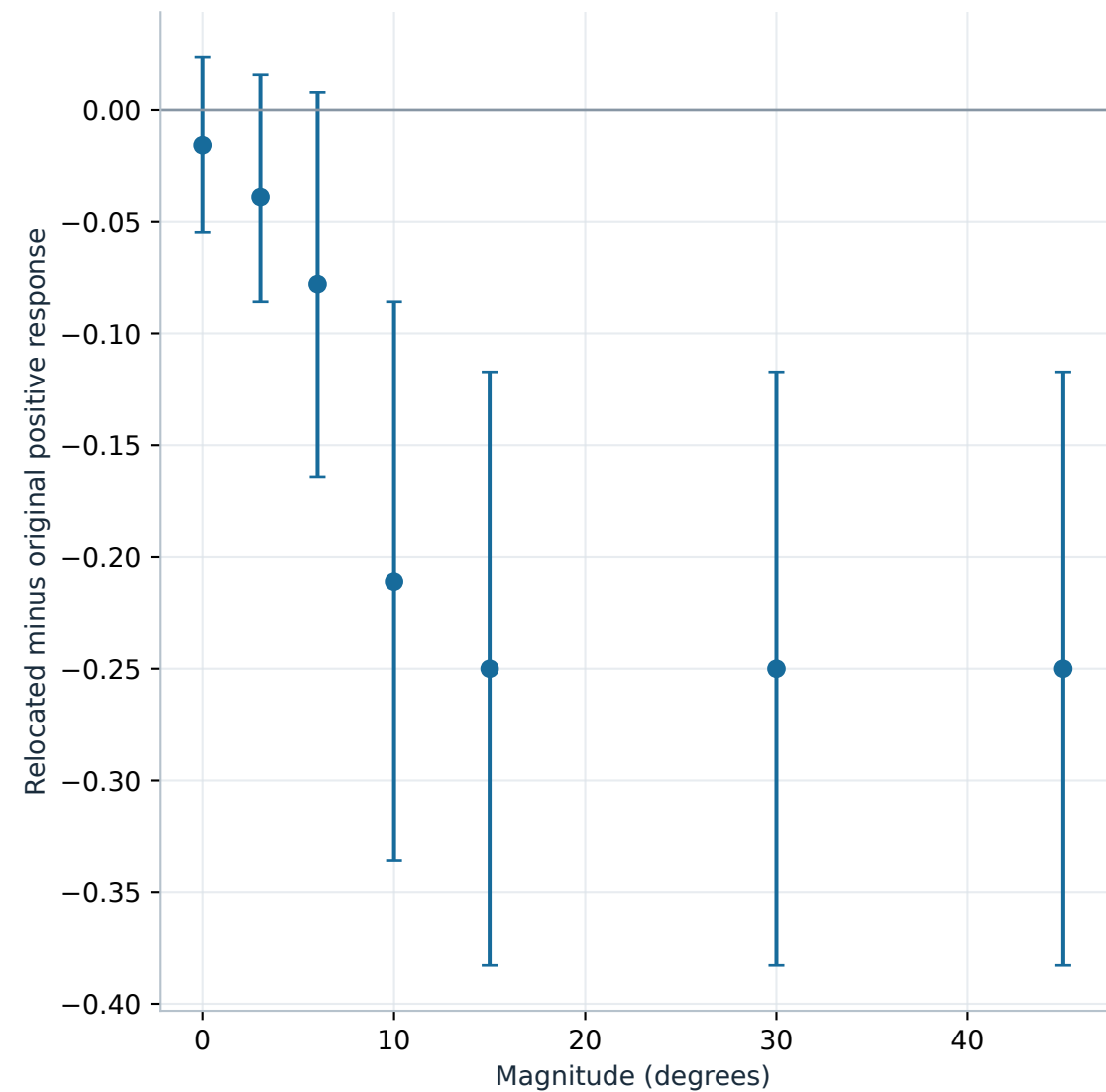
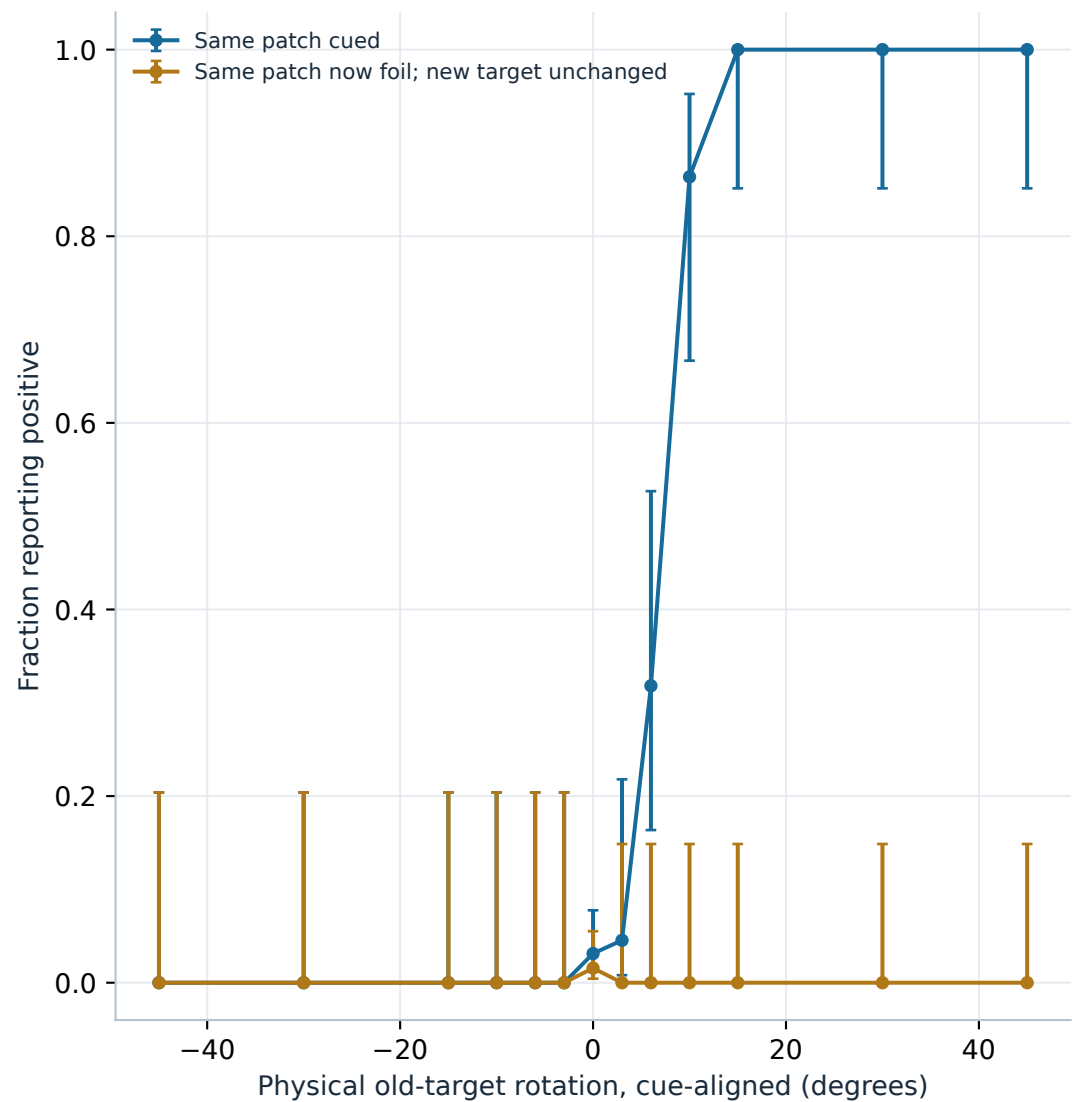


Does the observer use target rather than foil evidence?



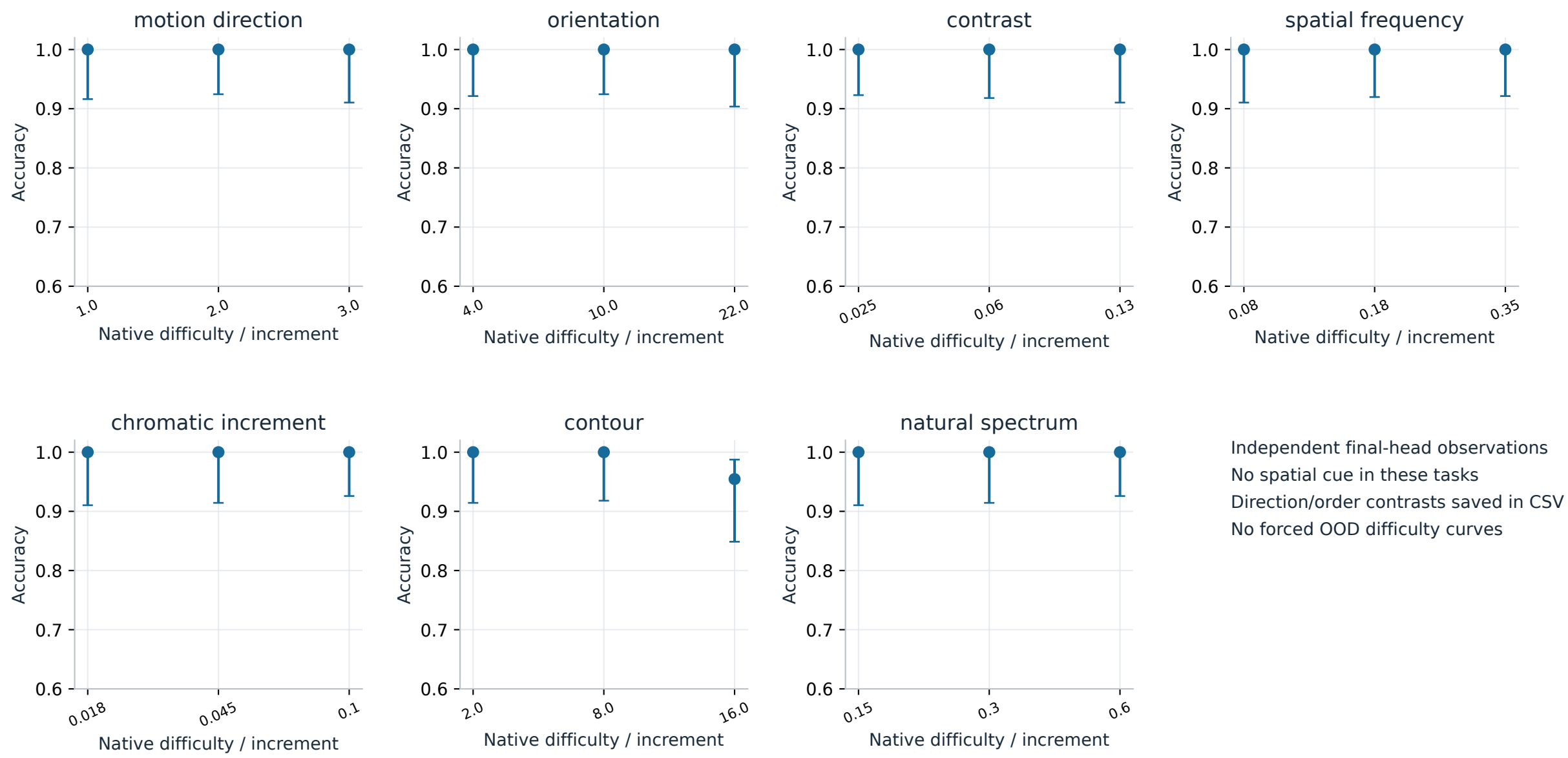
Empirical final positive choices with 95% Wilson intervals. Negative = opposite-sign or unchanged target. Blue: target rotation. Amber: designated next-location foil, conditioning target unchanged. Zero is the all-zero catch sweep only. 0/3/6/10 degrees are analysis-only OOD magnitudes; 15/30/45 are native. Native pattern couples foil rotations; these are conditional descriptions, not independently randomized evidence weights.

Matched sensory scenes; cue identity changes the requested report



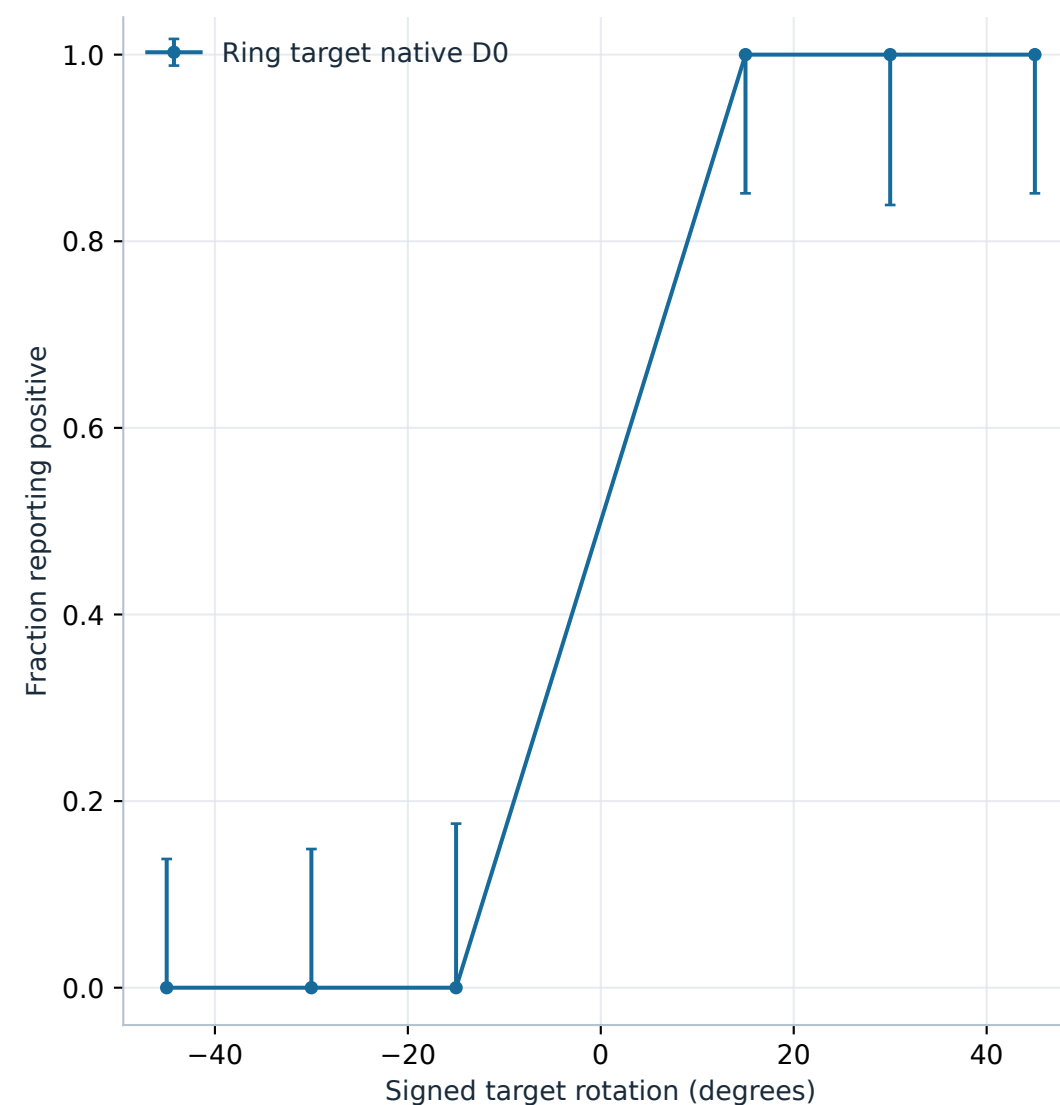
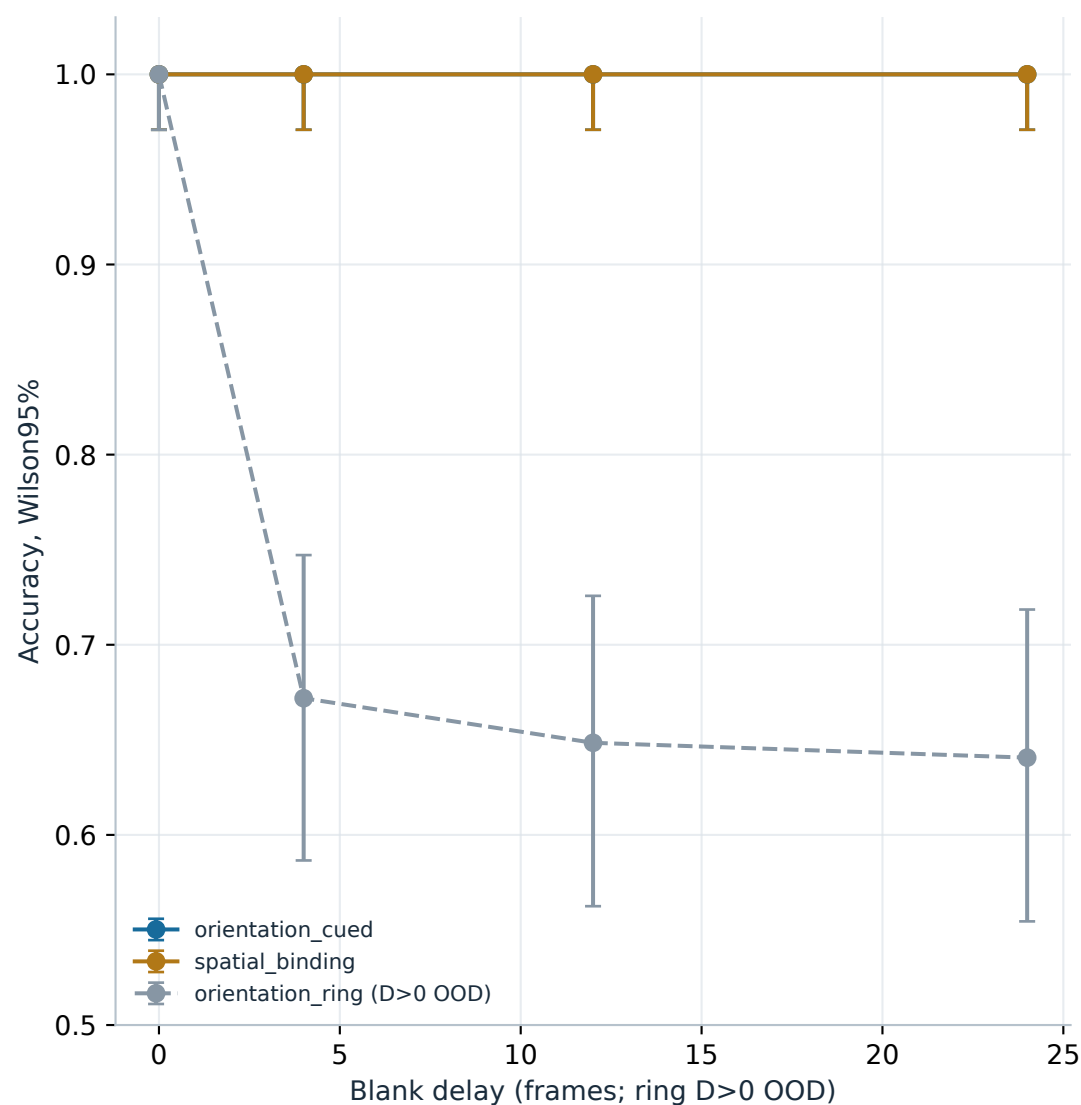
Only cue overlays change, with raw pre-overlay scenes and all rotations held fixed. New labels are recomputed. Left conditions the newly cued target to be unchanged; the same old-target physical evidence is now irrelevant. Right uses all matched pairs at each magnitude; 95% paired trial bootstrap. This is target reassignment, NOT an invalid-cue benefit or a cue-validity experiment.

Native sensory competence at checkpoint 35039



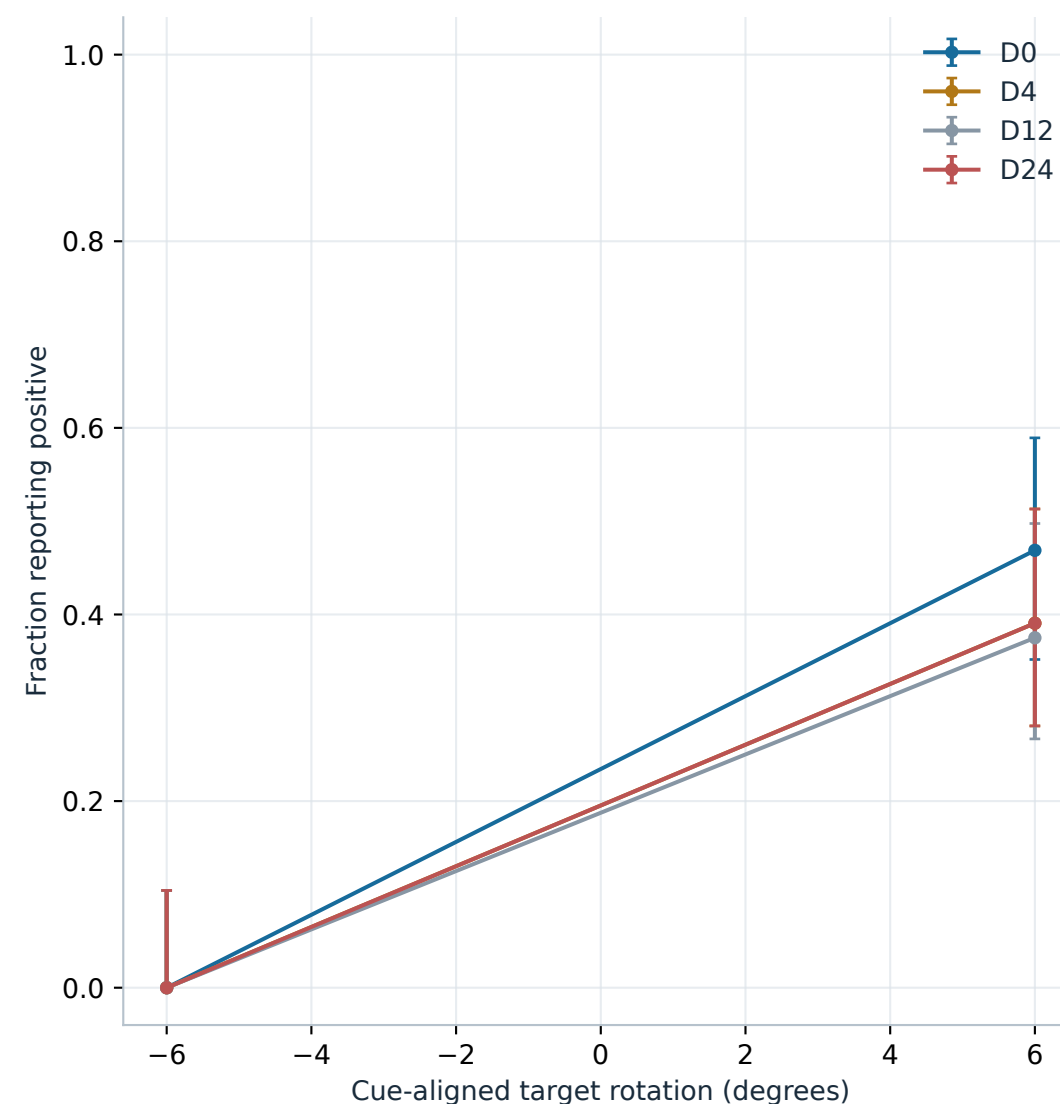
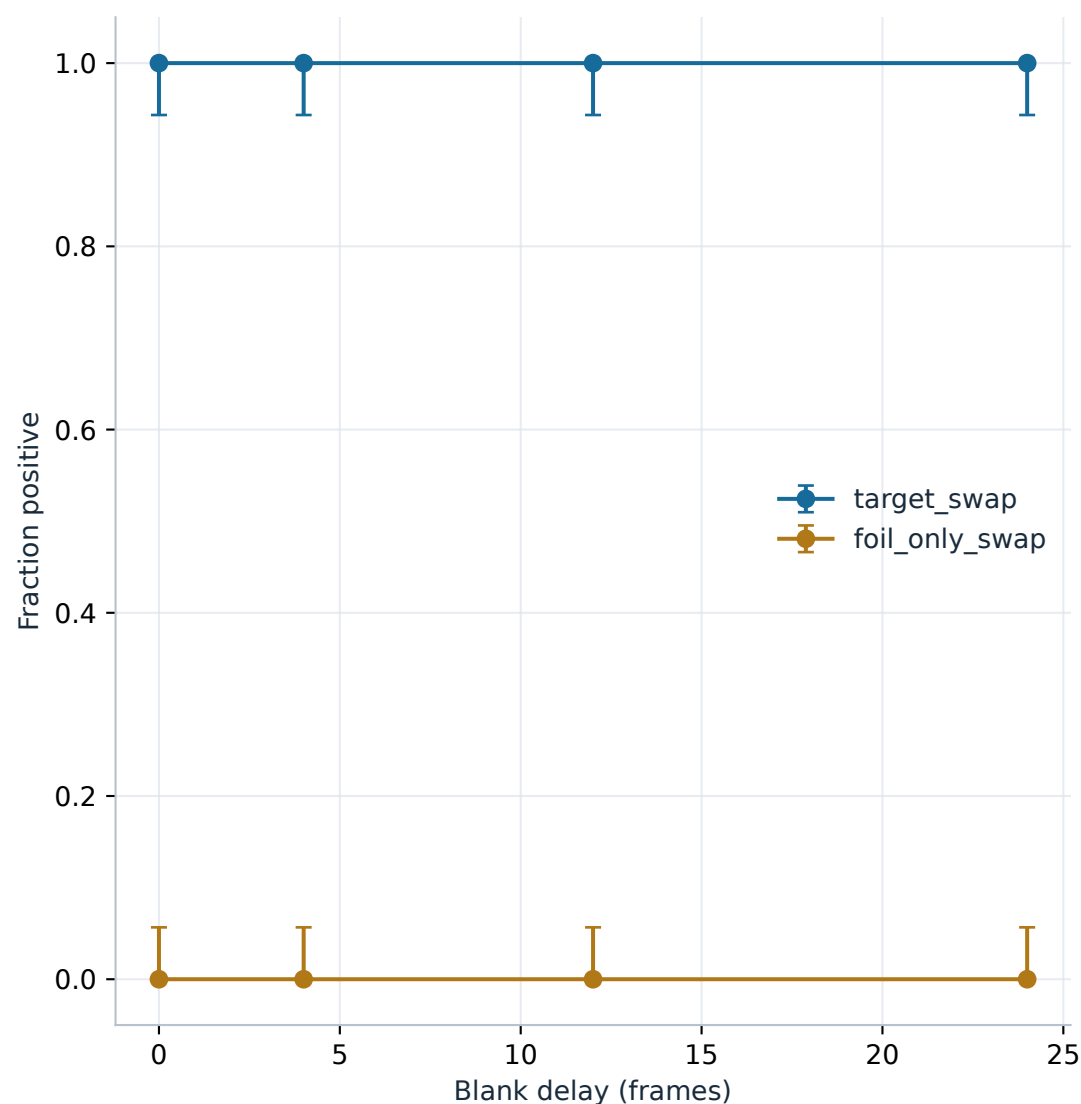
All seven sensory tasks use native generators and native difficulty draws. Points are empirical accuracy with Wilson 95% intervals and saved cell denominators. At a ceiling these data cannot identify thresholds or lapses. Binary interval labels mean frame0/frame1 for stronger evidence, not generic change/no-change. No cued/uncued allocation comparator is fabricated.

Retention and ring-cue response



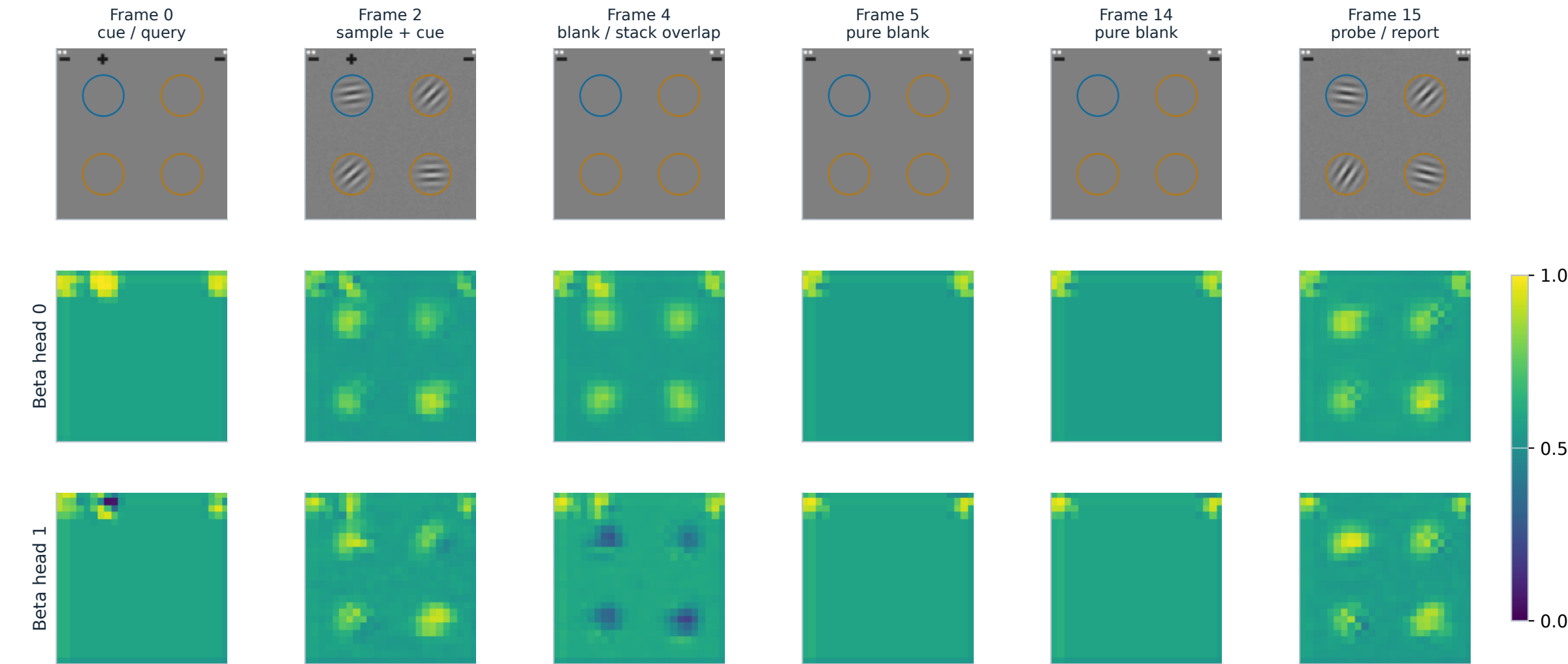
Cued orientation and binding: native D0/4/12/24, matched renderer seed across delays. Ring: ONLY D0 is trained; dashed D4/12/24 are supplemental OOD native-renderer delays, not solved-task validation. Right shows trained ring D0 signed rotation, without a sign glyph. First two nominal blanks overlap samples in stack-three inputs; only later blanks are stimulus-free. No decay constant is fitted at a ceiling.

Target swaps and below-native orientation difficulty



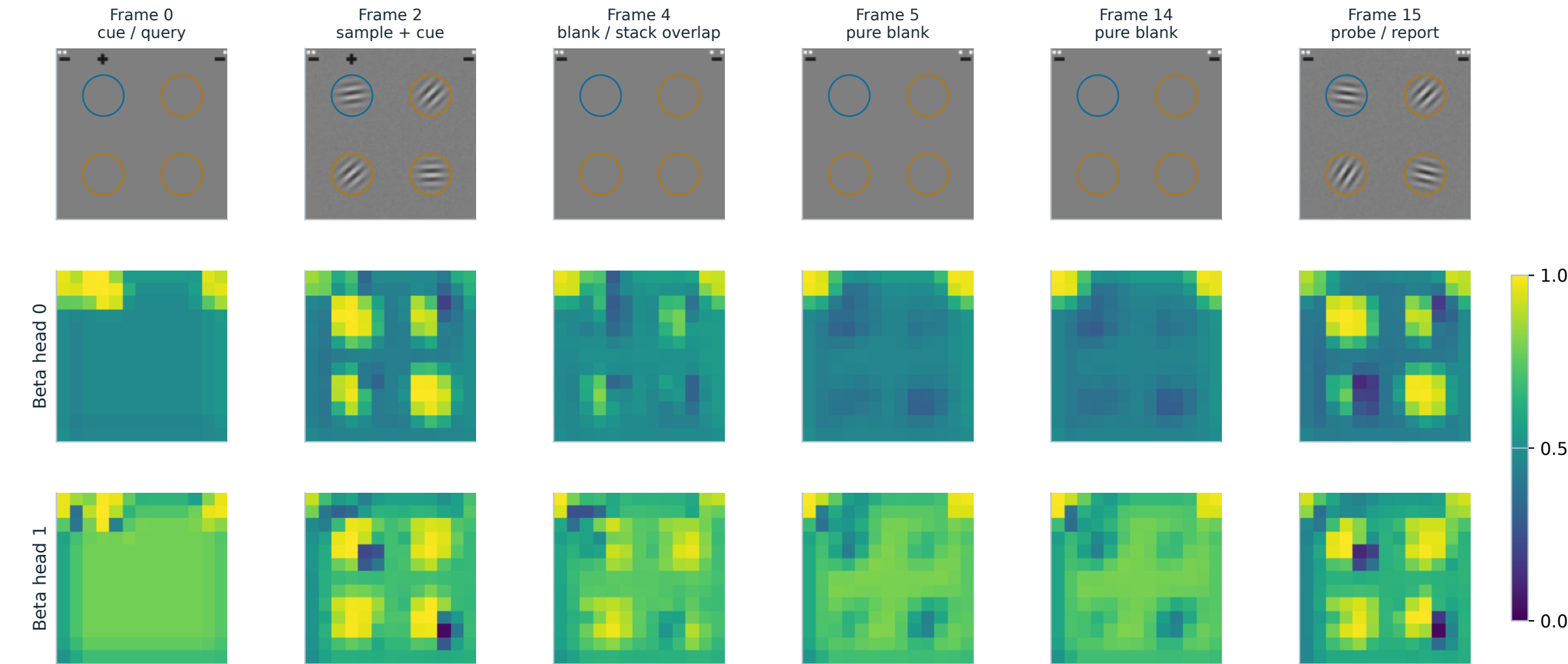
Binding requires only the subsequently queried location to swap; every trial contains two exchanged items. The query is AFTER retention, not a pre-cue. Right: 6-degree orientation is an explicit analysis-only OOD magnitude, useful for measuring non-ceiling behavior without modifying training. Curves are observed choices, not fitted thresholds.

orientation cued — real trial, scale 0



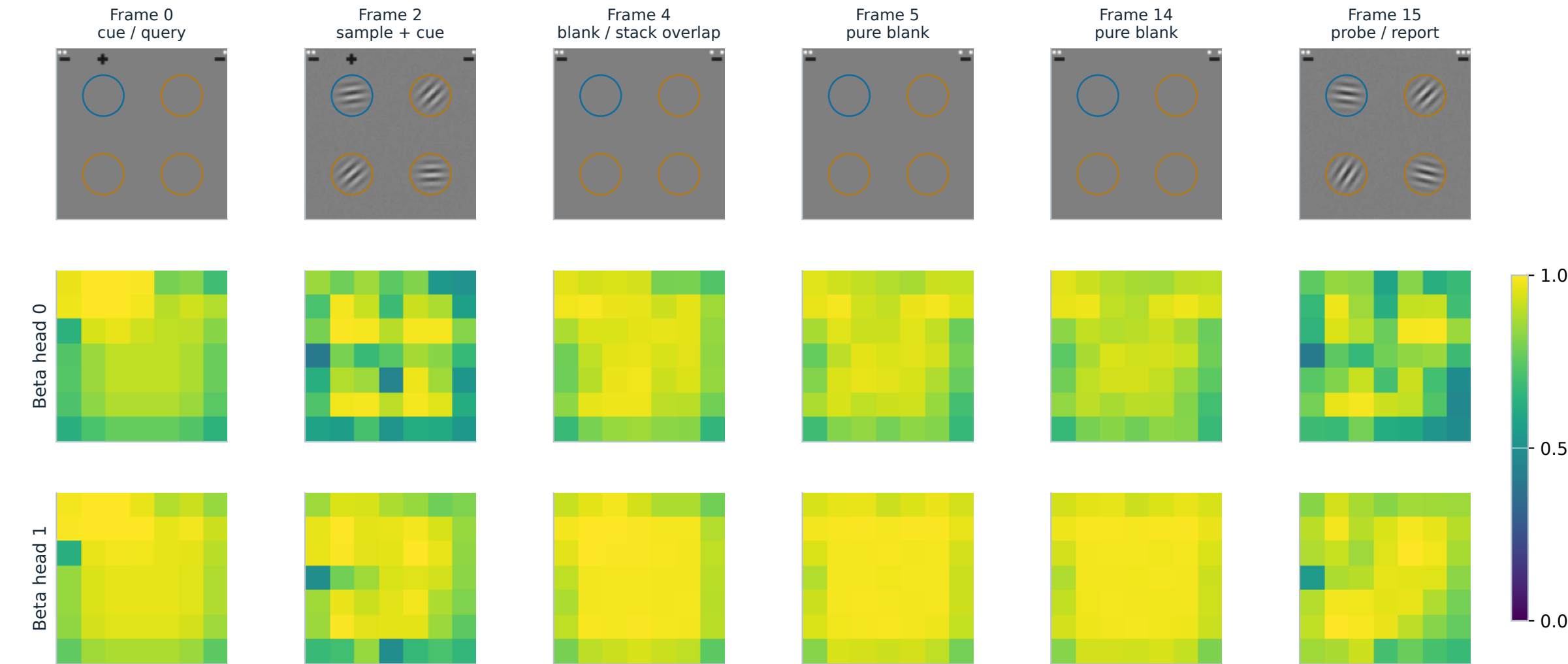
Trial0, not a trial average. Top: actual scene, blue target/eventual-query site and amber foils. Bottom: head-separated beta write/correction gates, fixed scale0-1. Maps cover actual 25/13/7 grids, not reconstructed regional values. Frame4 remains stack-overlap; frame5 is first pure blank. Every trial/frame and alpha/coefficient/ConvGRU map is selectable offline. Binding target is unknown to observer until query.

orientation cued — real trial, scale 1



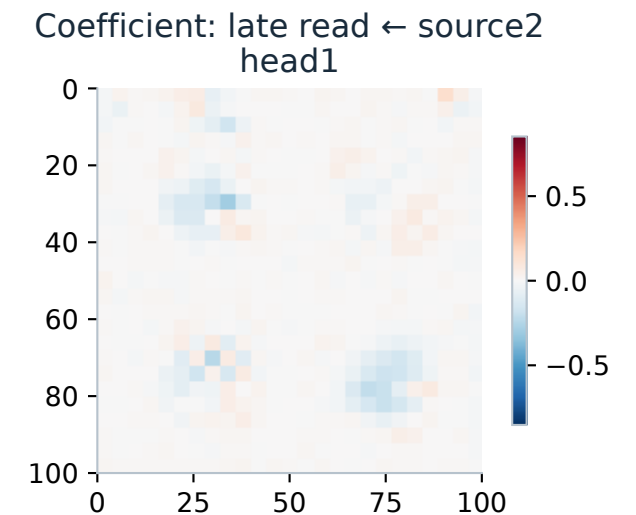
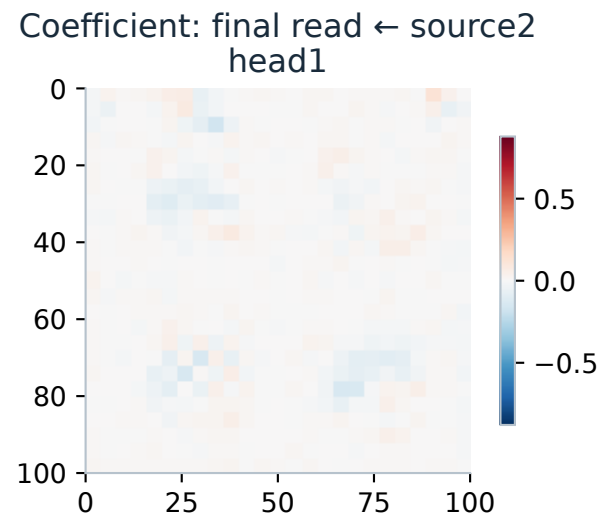
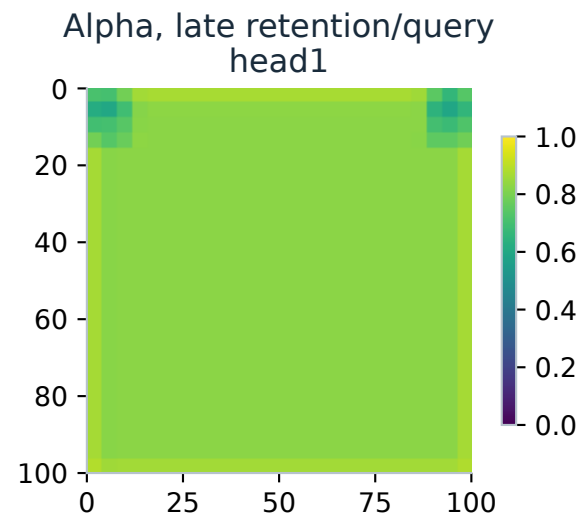
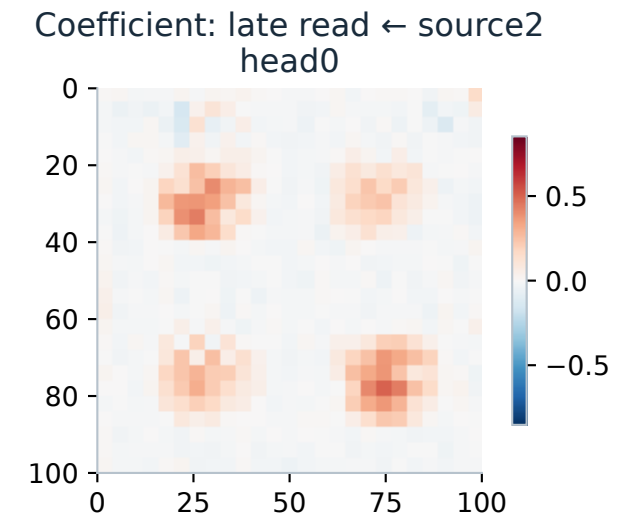
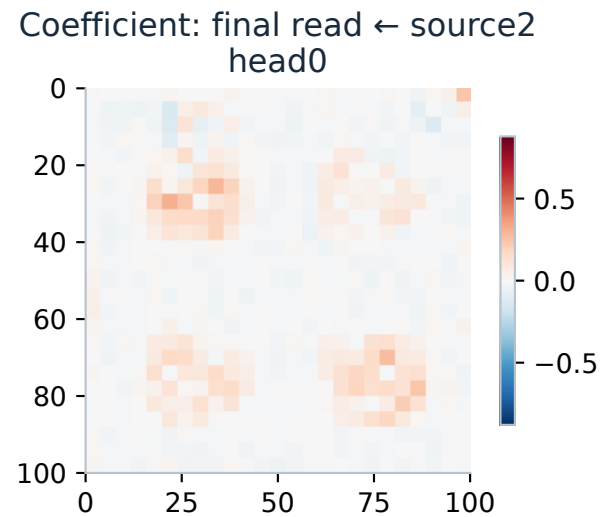
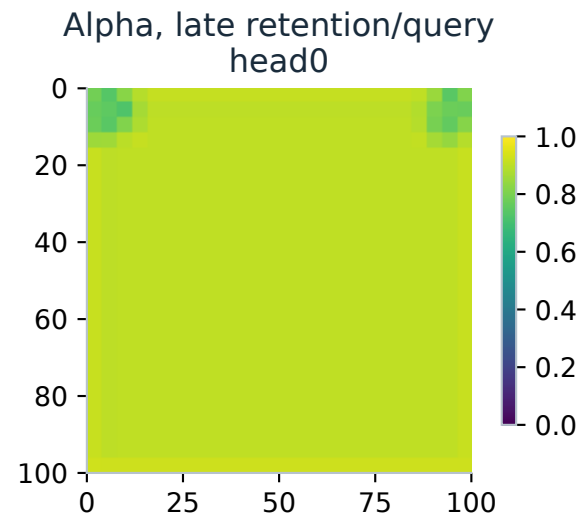
Trial0, not a trial average. Top: actual scene, blue target/eventual-query site and amber foils. Bottom: head-separated beta write/correction gates, fixed scale0-1. Maps cover actual 25/13/7 grids, not reconstructed regional values. Frame4 remains stack-overlap; frame5 is first pure blank. Every trial/frame and alpha/coefficient/ConvGRU map is selectable offline. Binding target is unknown to observer until query.

orientation cued — real trial, scale 2



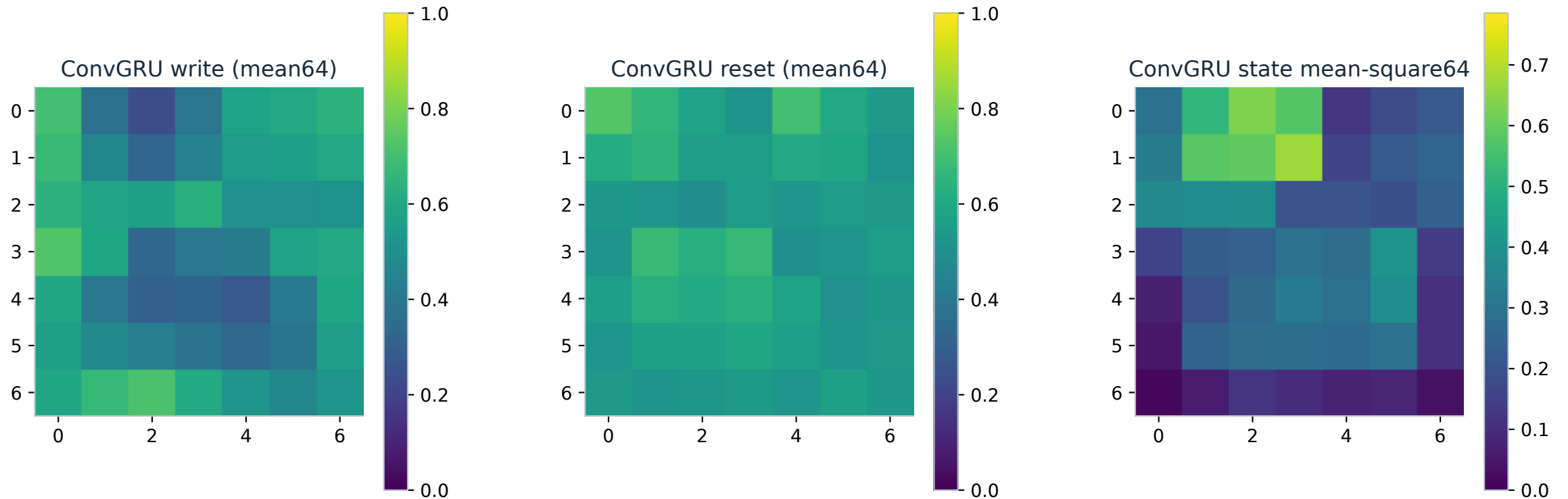
Trial0, not a trial average. Top: actual scene, blue target/eventual-query site and amber foils. Bottom: head-separated beta write/correction gates, fixed scale0-1. Maps cover actual 25/13/7 grids, not reconstructed regional values. Frame4 remains stack-overlap; frame5 is first pure blank. Every trial/frame and alpha/coefficient/ConvGRU map is selectable offline. Binding target is unknown to observer until query.

Retention gates and signed local temporal coefficients



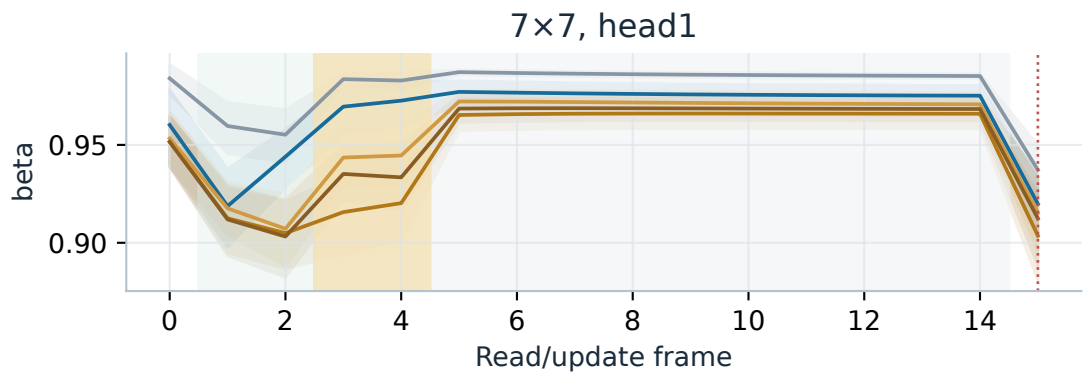
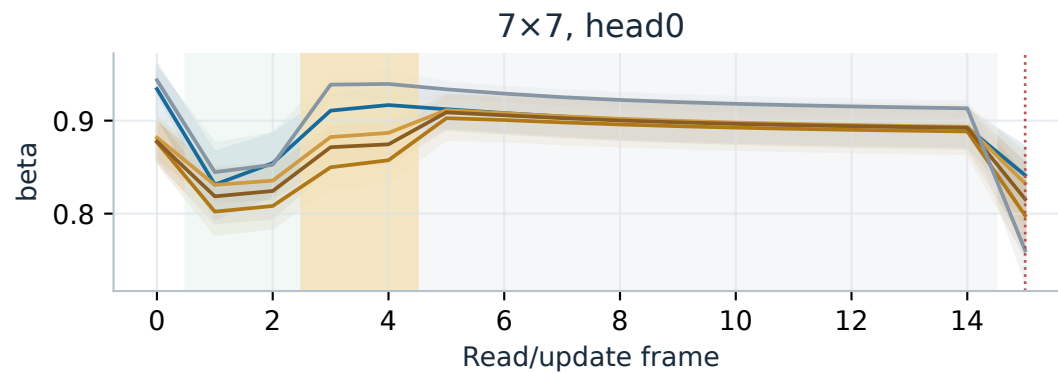
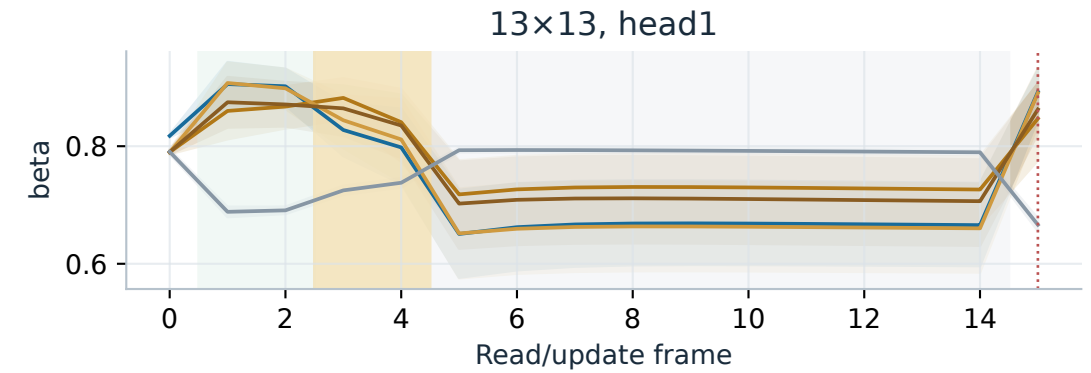
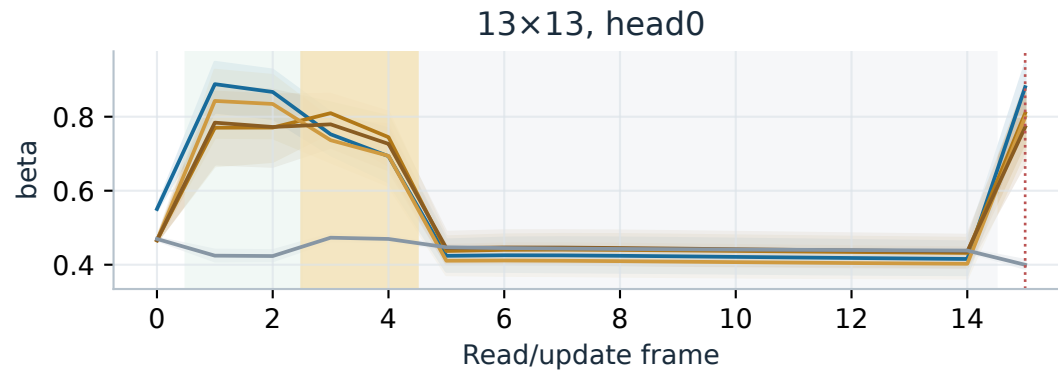
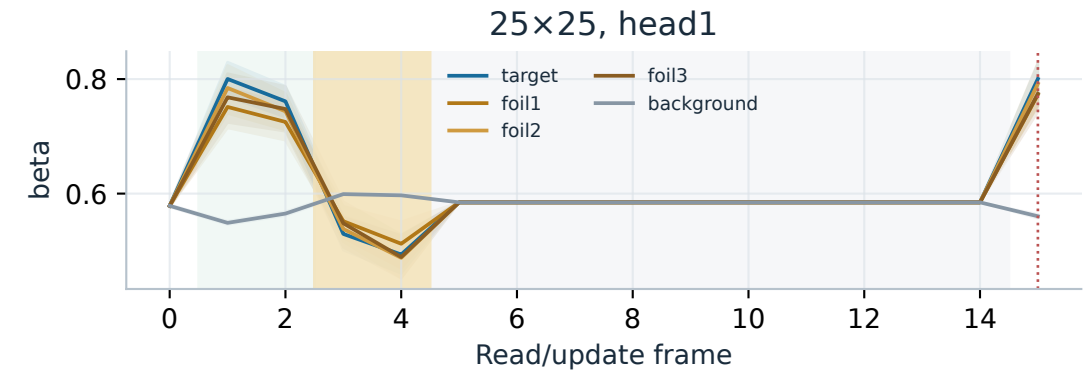
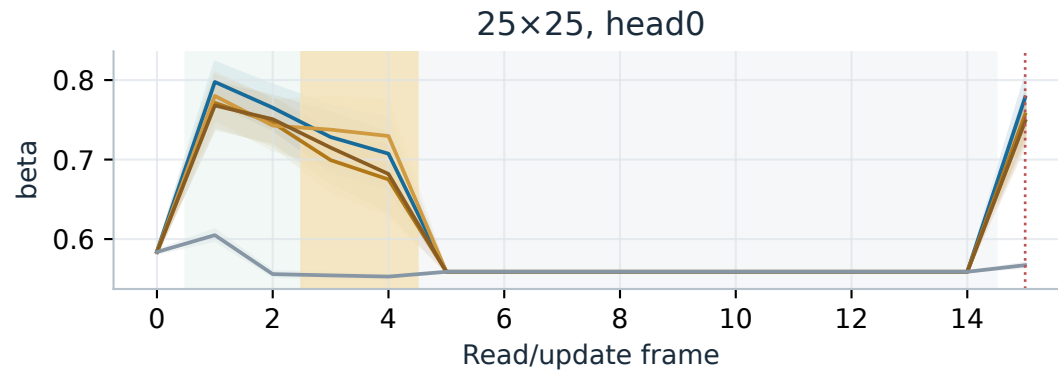
Actual trial0 arrays. Alpha is the mean over 8 retention rows WITHIN each separate head. Signed coefficients can be negative and are neither spatial softmax probabilities nor final-decision attribution. Final-read arrays retain every source frame; fixed-source2 arrays retain every read frame. Stable range per quantity across saved trials/times. Exact weighted-value reconstruction was verified before output convolution.

Downstream 7×7 recurrent readout — distinct measurements



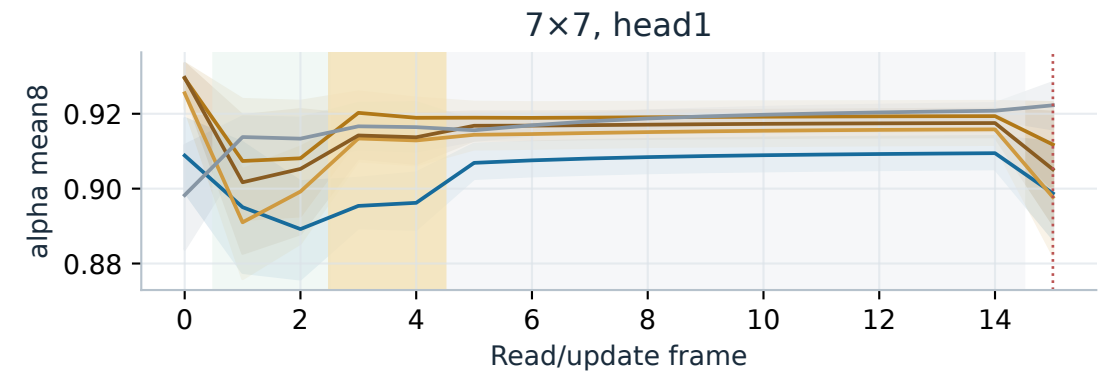
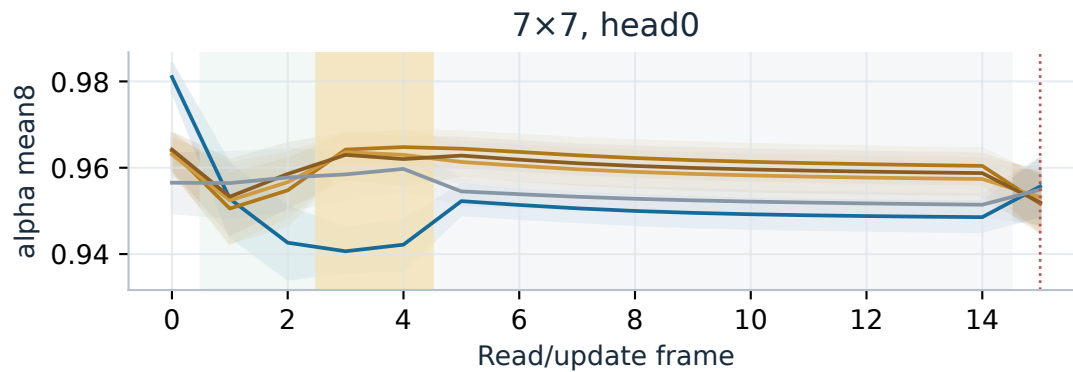
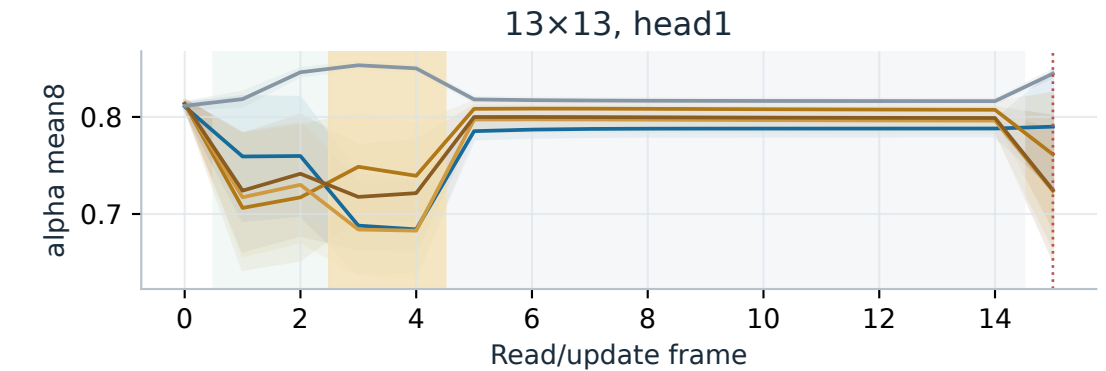
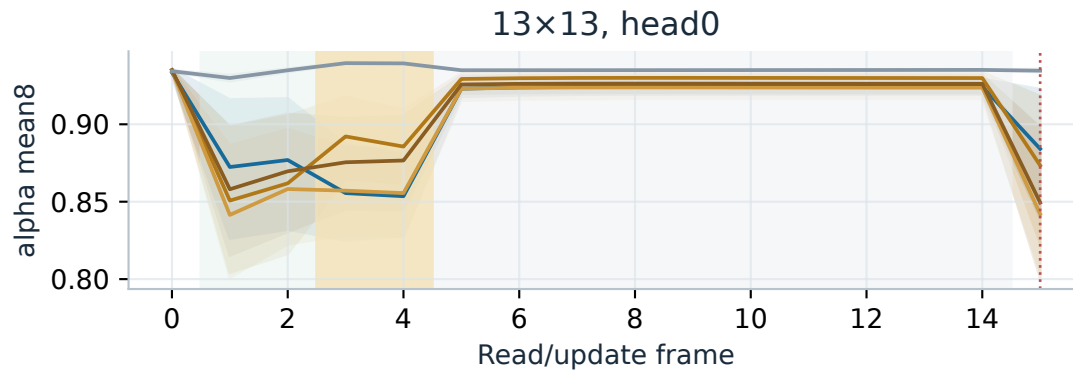
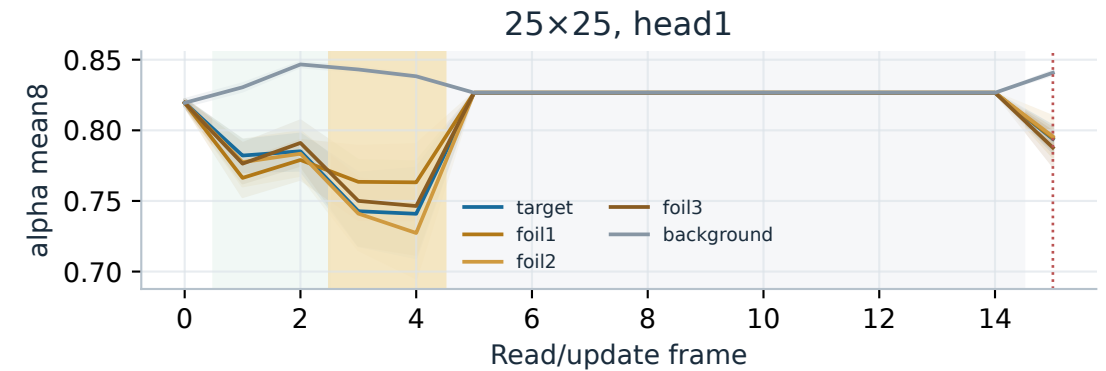
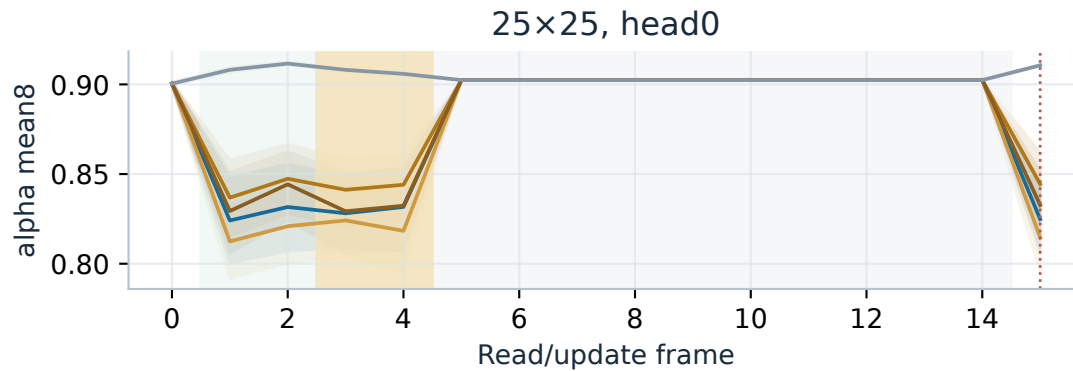
Actual trial0 at frame14. These are ConvGRU write/reset gates and state energy, not KDA gates, implicit coefficients or attention probabilities. Channel averaging is stated explicitly. Full per-timestep arrays retained; no intermediate head is scored as an in-distribution decision.

orientation cued — beta allocation



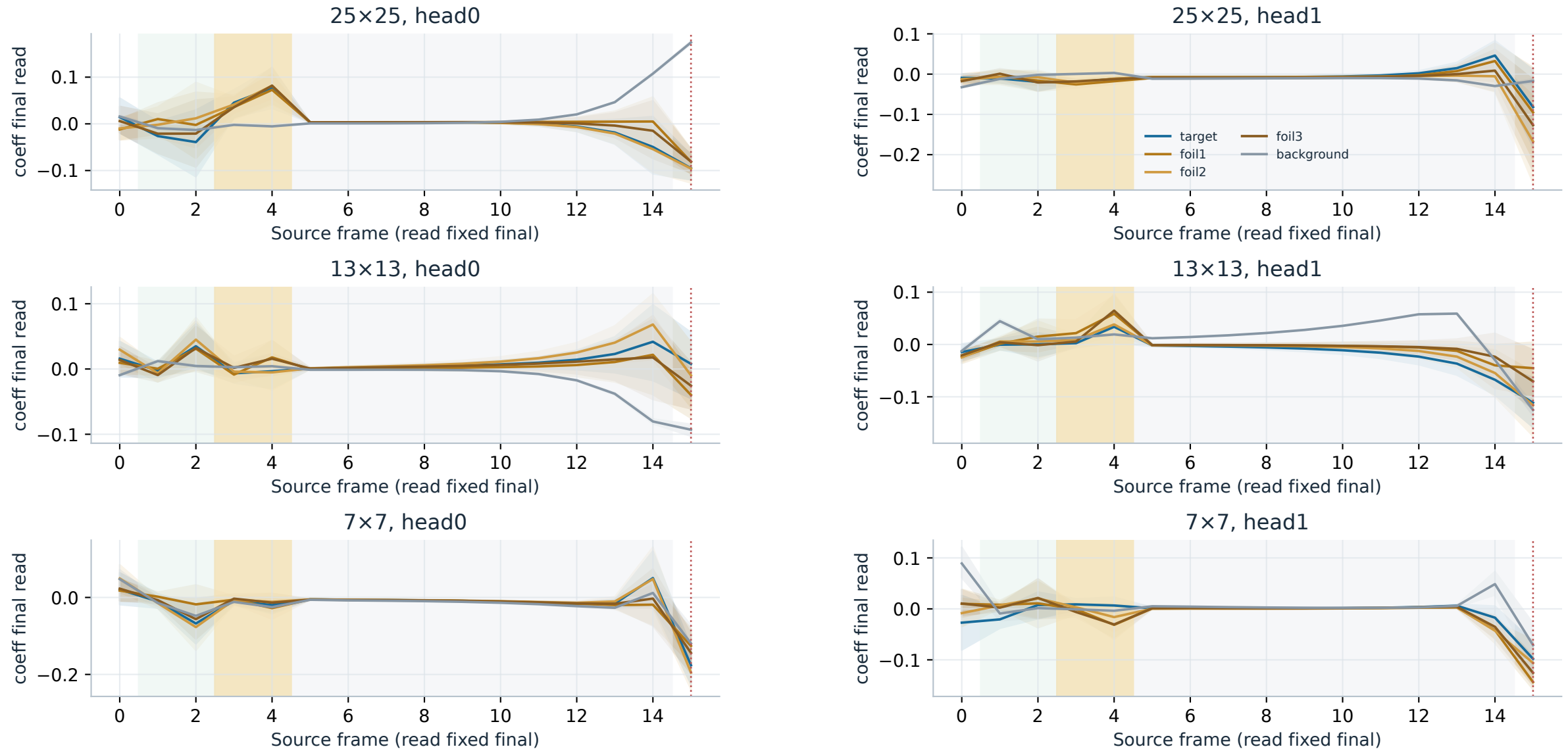
16 independent native D12 trials; each foil separate. Matched translated kernels have equal support/mass at each scale; plotted means are per-mask-mass, not regional totals. Shading is 95% trial-bootstrap uncertainty, never pixels/heads as replicates. Amber phase marks two stack-overlap blanks. For binding, target is only a retrospective eventual-query grouping before the query. Signed coefficient panel preserves sign; separate mean-absolute values are in source CSV.

orientation cued — alpha mean8 allocation



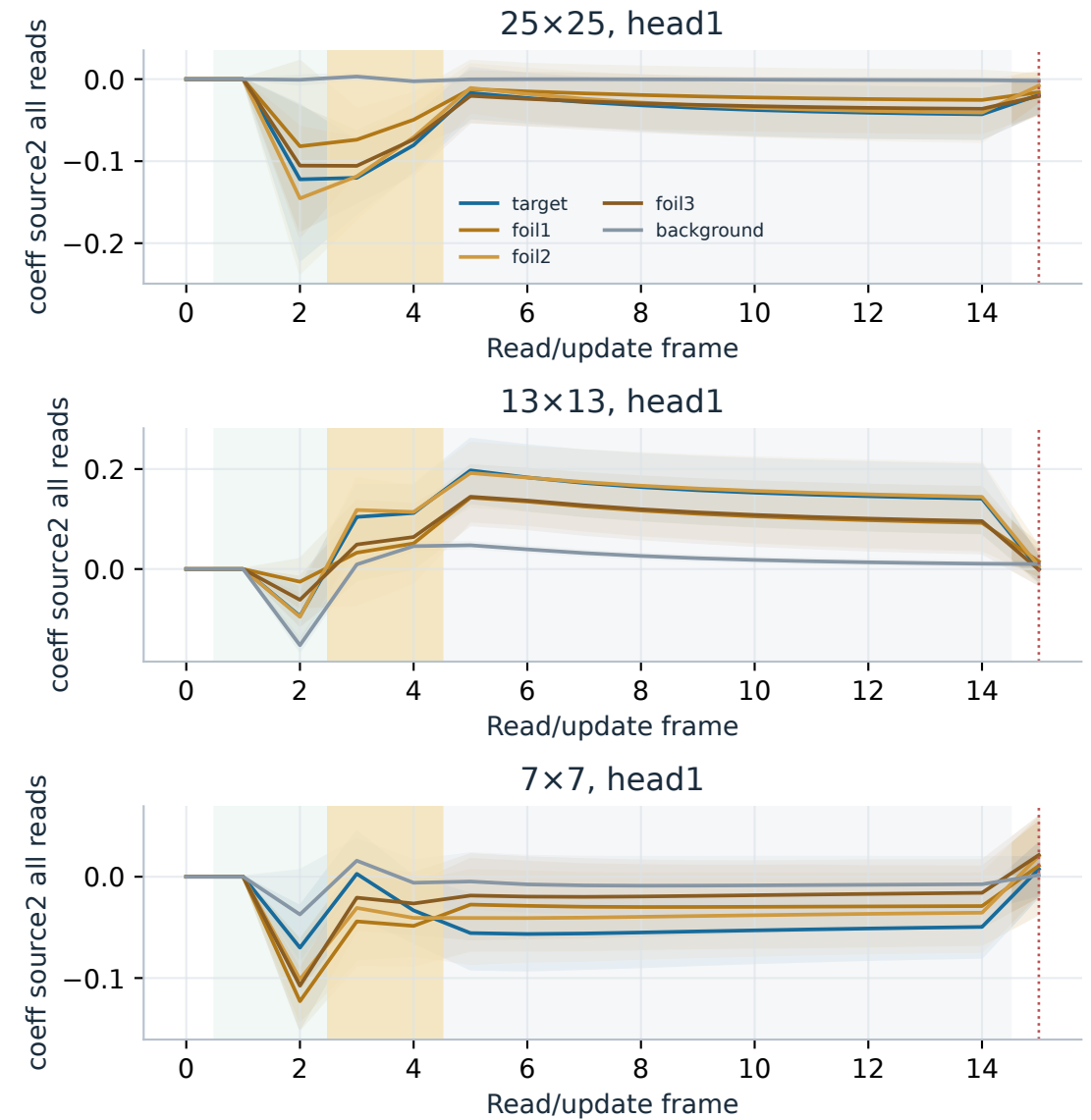
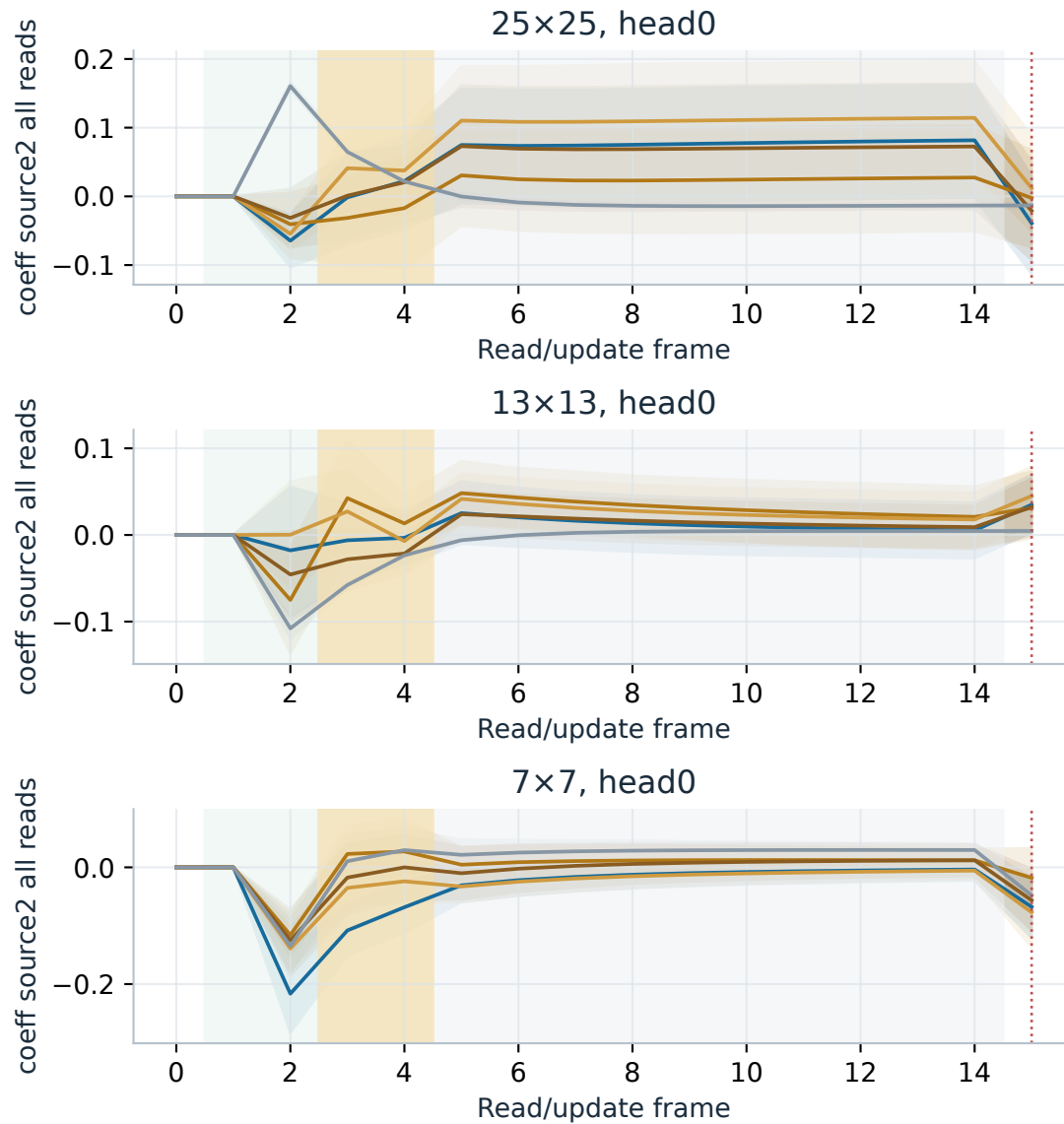
16 independent native D12 trials; each foil separate. Matched translated kernels have equal support/mass at each scale; plotted means are per-mask-mass, not regional totals. Shading is 95% trial-bootstrap uncertainty, never pixels/heads as replicates. Amber phase marks two stack-overlap blanks. For binding, target is only a retrospective eventual-query grouping before the query. Signed coefficient panel preserves sign; separate mean-absolute values are in source CSV.

orientation cued — coeff final read allocation



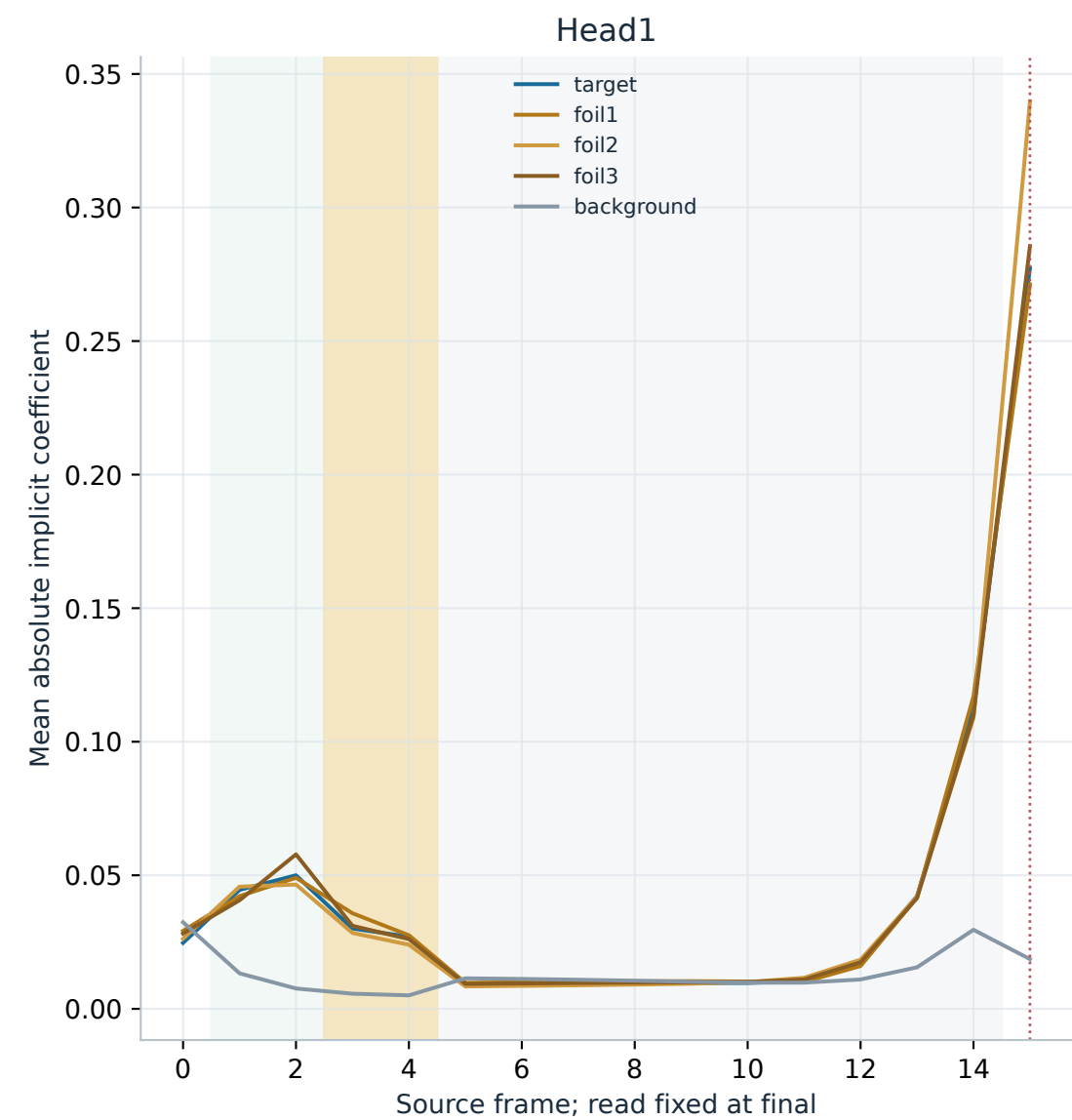
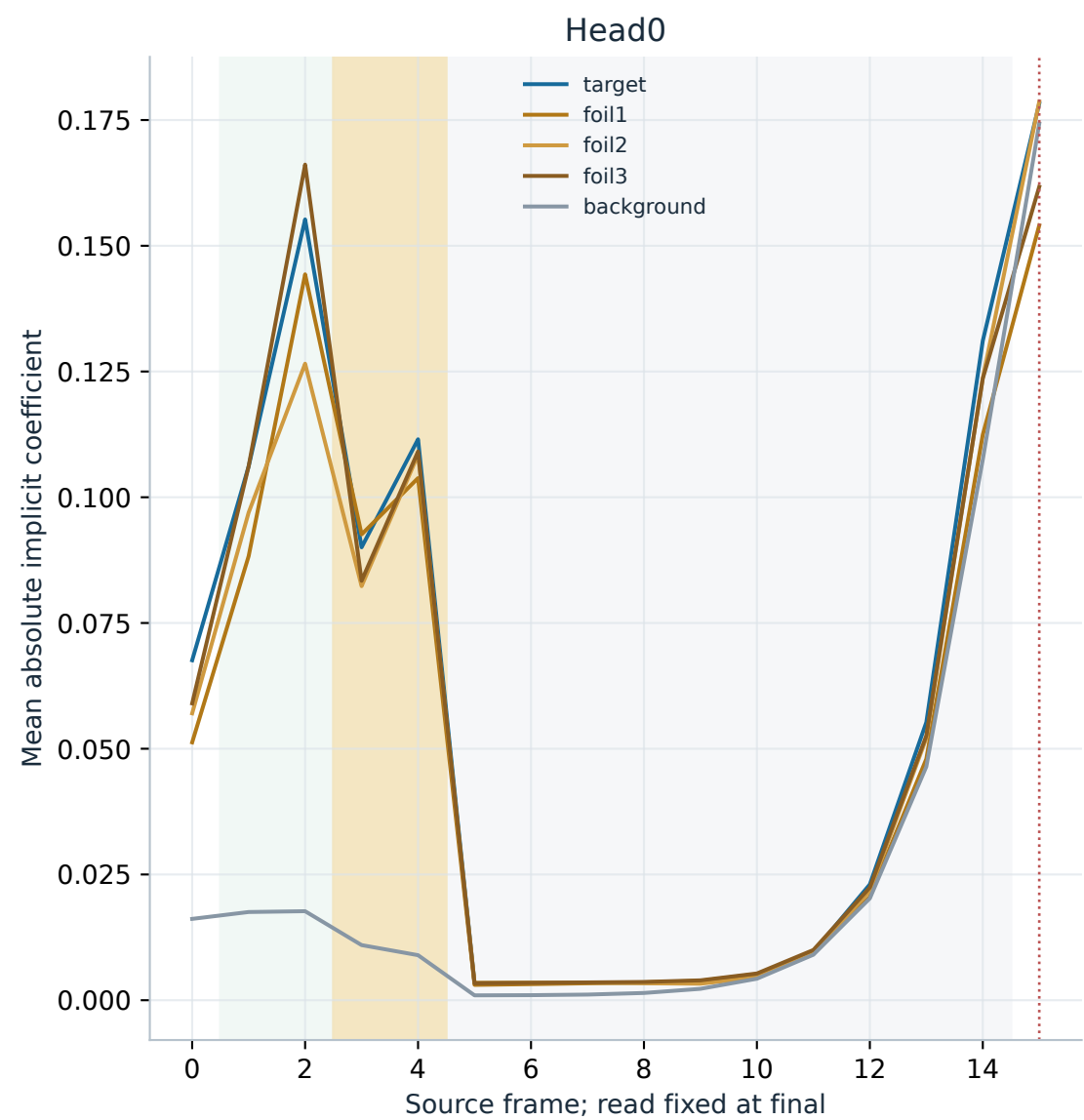
16 independent native D12 trials; each foil separate. Matched translated kernels have equal support/mass at each scale; plotted means are per-mask-mass, not regional totals. Shading is 95% trial-bootstrap uncertainty, never pixels/heads as replicates. Amber phase marks two stack-overlap blanks. For binding, target is only a retrospective eventual-query grouping before the query. Signed coefficient panel preserves sign; separate mean-absolute values are in source CSV.

orientation cued — coeff source2 all reads allocation



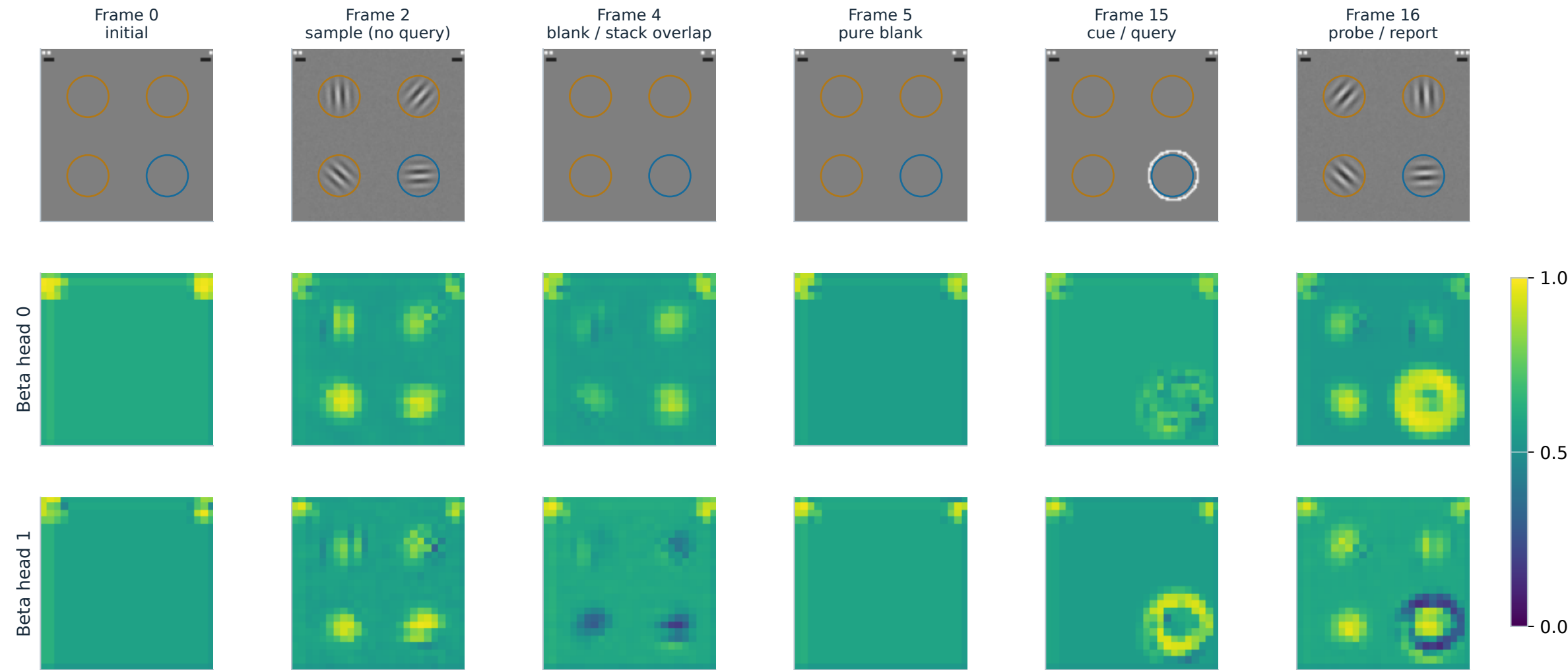
16 independent native D12 trials; each foil separate. Matched translated kernels have equal support/mass at each scale; plotted means are per-mask-mass, not regional totals. Shading is 95% trial-bootstrap uncertainty, never pixels/heads as replicates. Amber phase marks two stack-overlap blanks. For binding, target is only a retrospective eventual-query grouping before the query. Signed coefficient panel preserves sign; separate mean-absolute values are in source CSV.

Absolute coefficient magnitude is not signed allocation



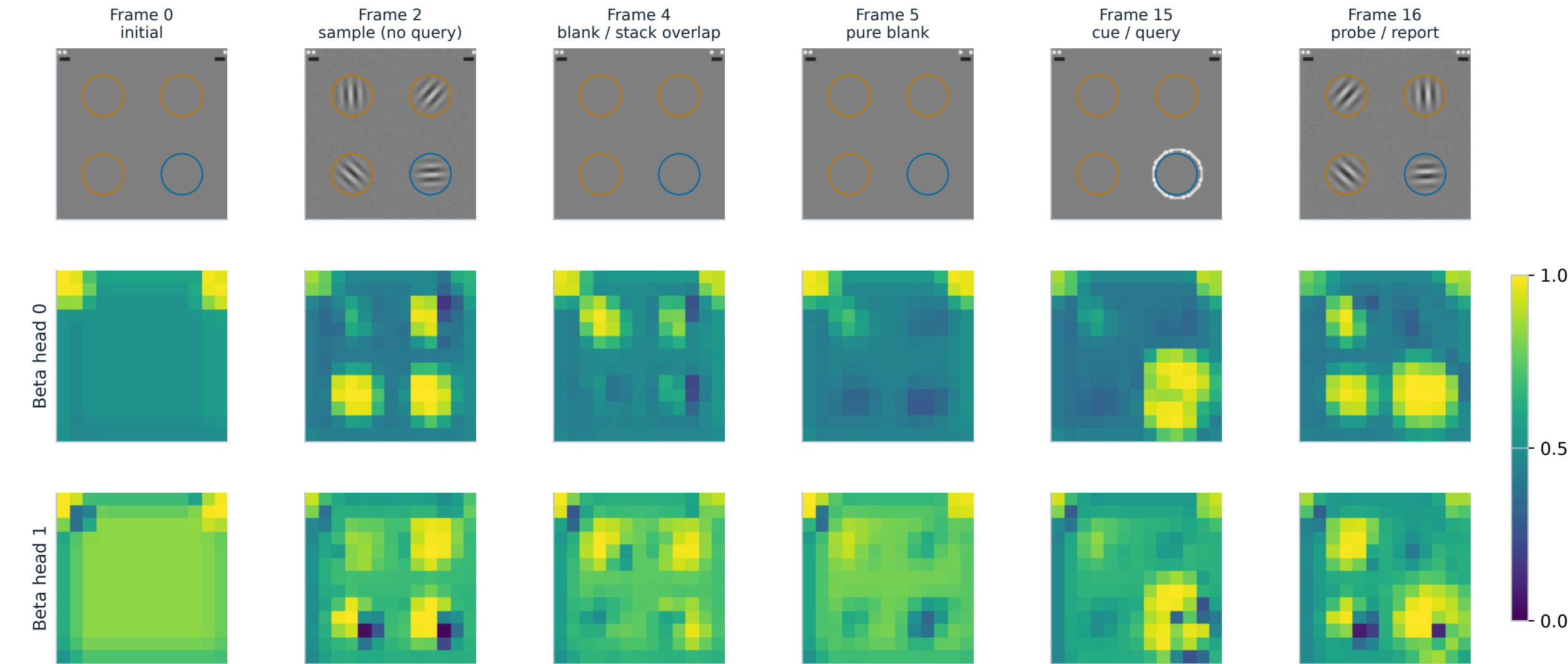
Mean of absolute local coefficients (not absolute of the signed mean). Heads remain separate; source time is not read time. The corresponding signed maps and all-read source2 maps remain in raw arrays and the offline viewer. These quantities describe local memory reads, not patch-to-patch attention or behavioral importance.

spatial binding — real trial, scale 0



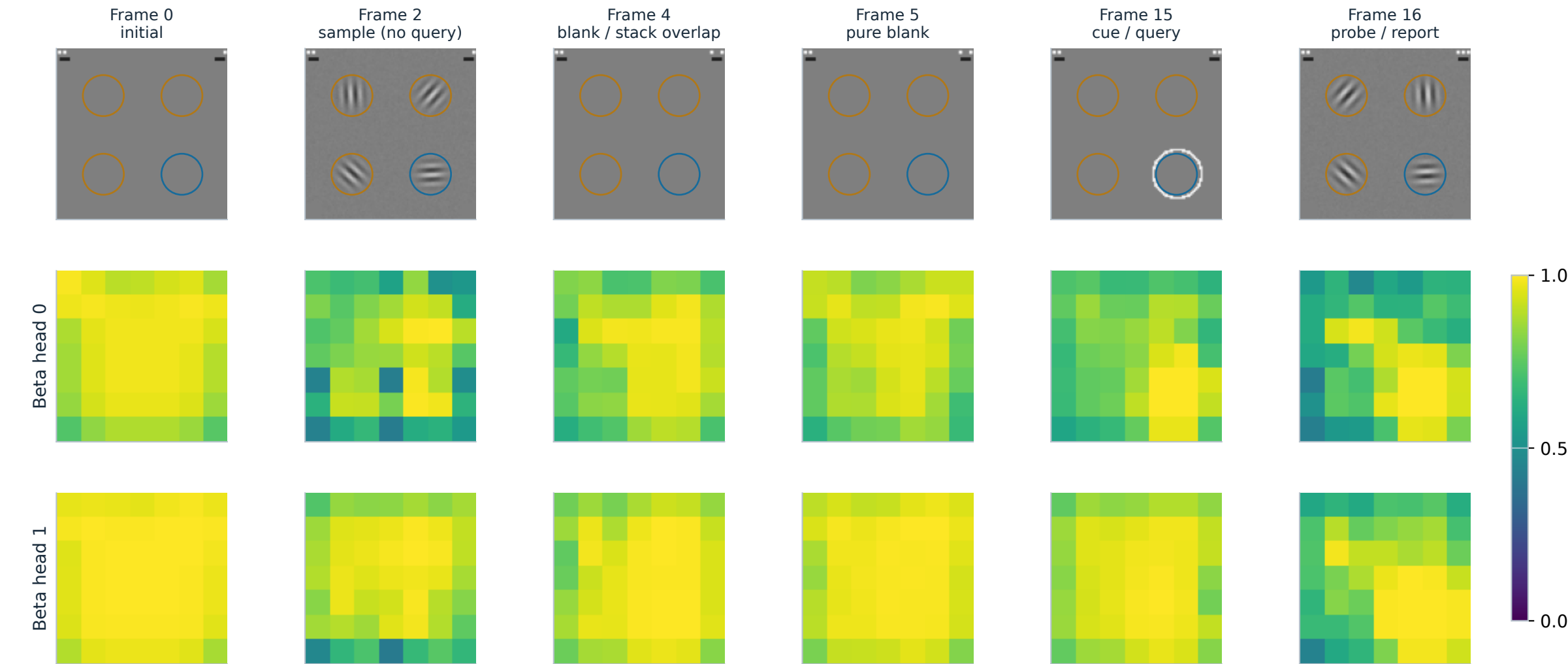
Trial0, not a trial average. Top: actual scene, blue target/eventual-query site and amber foils. Bottom: head-separated beta write/correction gates, fixed scale0-1. Maps cover actual 25/13/7 grids, not reconstructed regional values. Frame4 remains stack-overlap; frame5 is first pure blank. Every trial/frame and alpha/coefficient/ConvGRU map is selectable offline. Binding target is unknown to observer until query.

spatial binding — real trial, scale 1



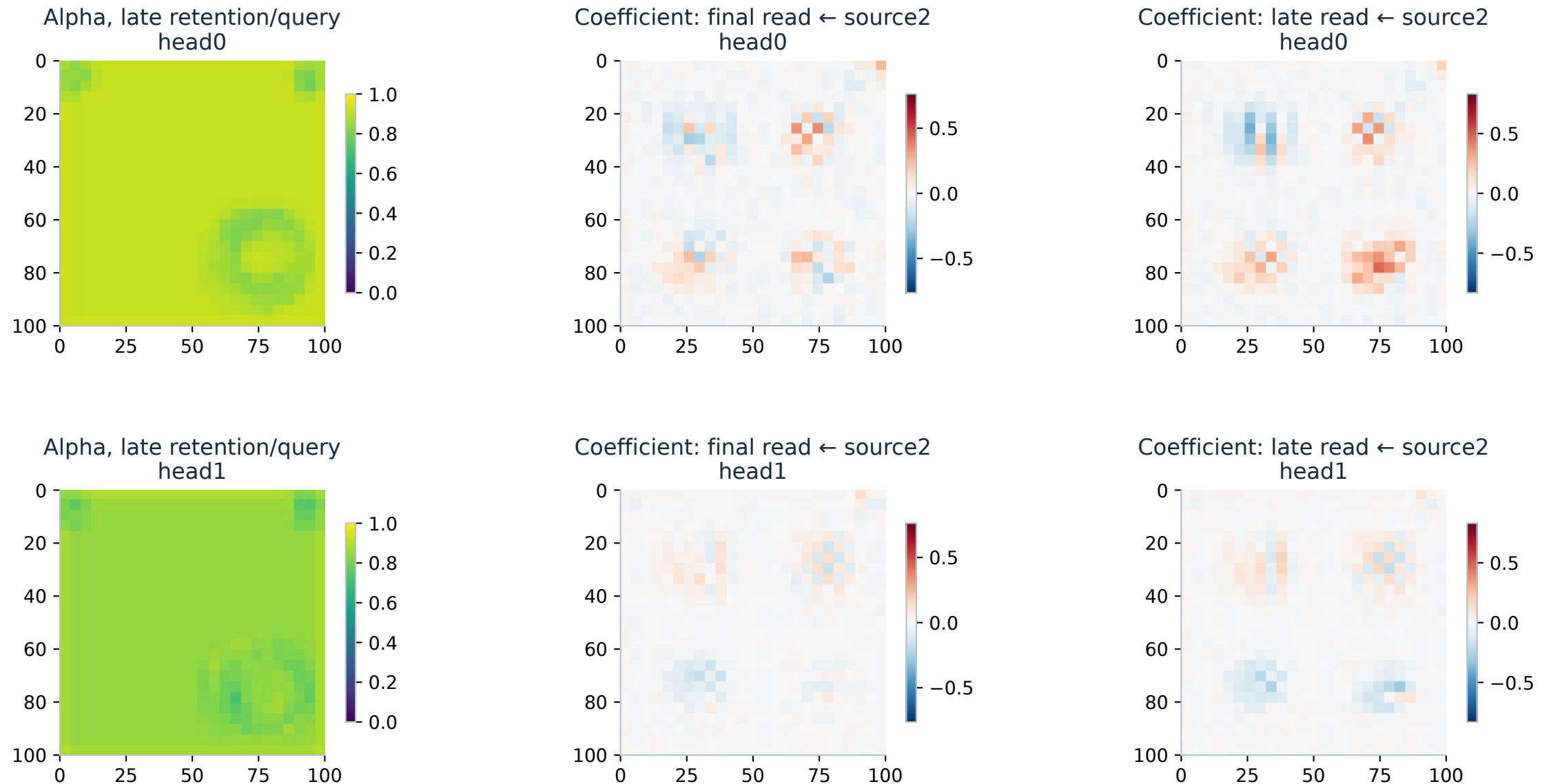
Trial0, not a trial average. Top: actual scene, blue target/eventual-query site and amber foils. Bottom: head-separated beta write/correction gates, fixed scale0-1. Maps cover actual 25/13/7 grids, not reconstructed regional values. Frame4 remains stack-overlap; frame5 is first pure blank. Every trial/frame and alpha/coefficient/ConvGRU map is selectable offline. Binding target is unknown to observer until query.

spatial binding — real trial, scale 2



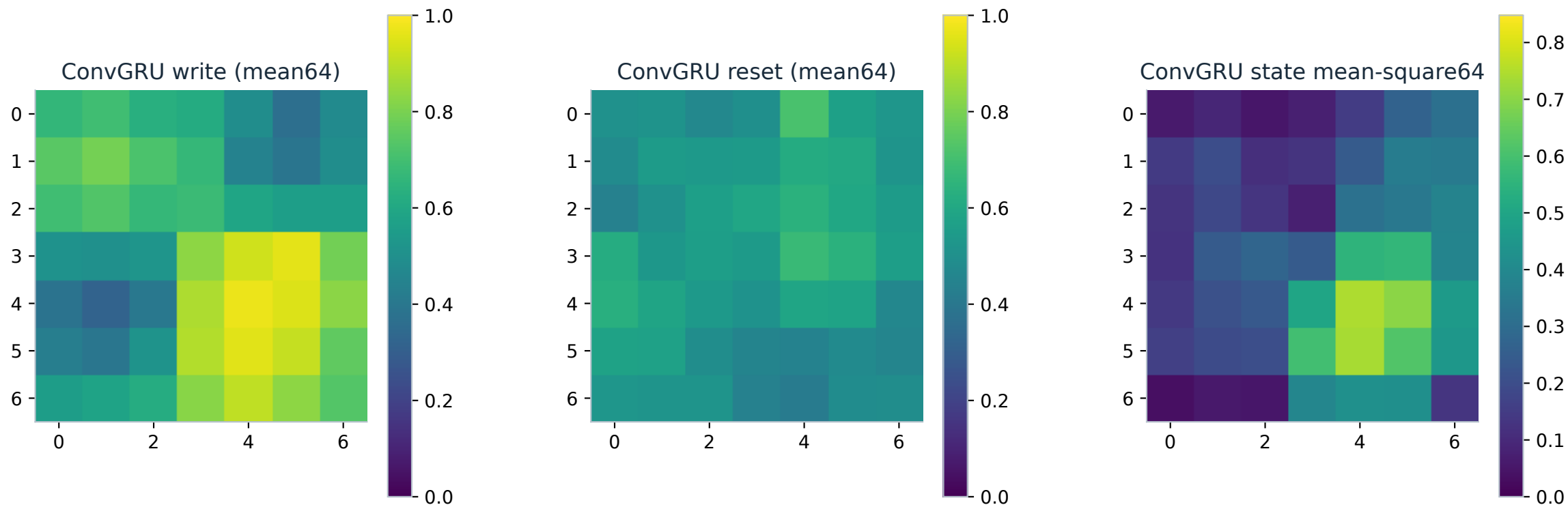
Trial0, not a trial average. Top: actual scene, blue target/eventual-query site and amber foils. Bottom: head-separated beta write/correction gates, fixed scale0-1. Maps cover actual 25/13/7 grids, not reconstructed regional values. Frame4 remains stack-overlap; frame5 is first pure blank. Every trial/frame and alpha/coefficient/ConvGRU map is selectable offline. Binding target is unknown to observer until query.

Retention gates and signed local temporal coefficients



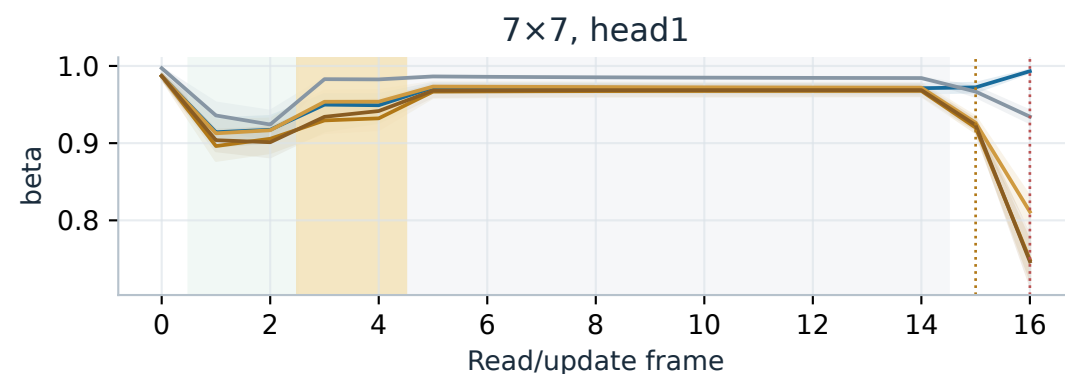
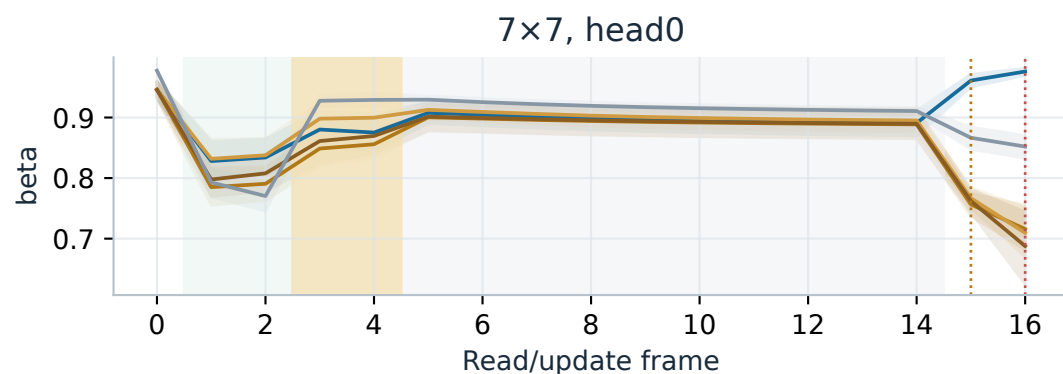
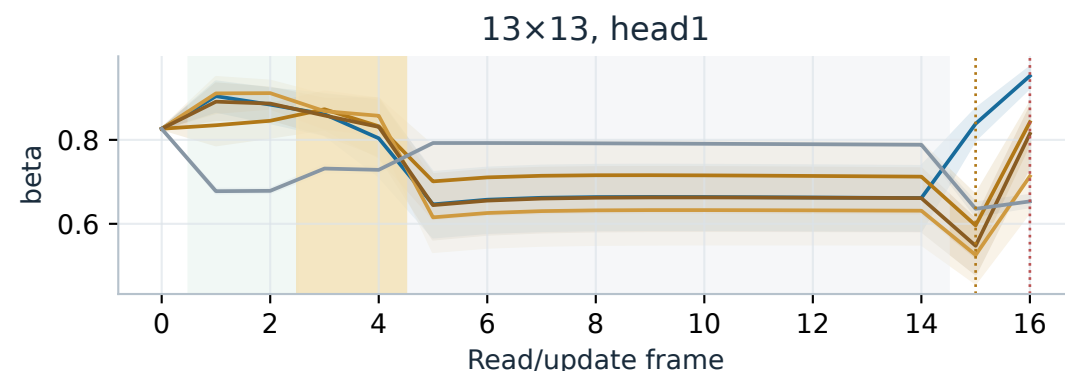
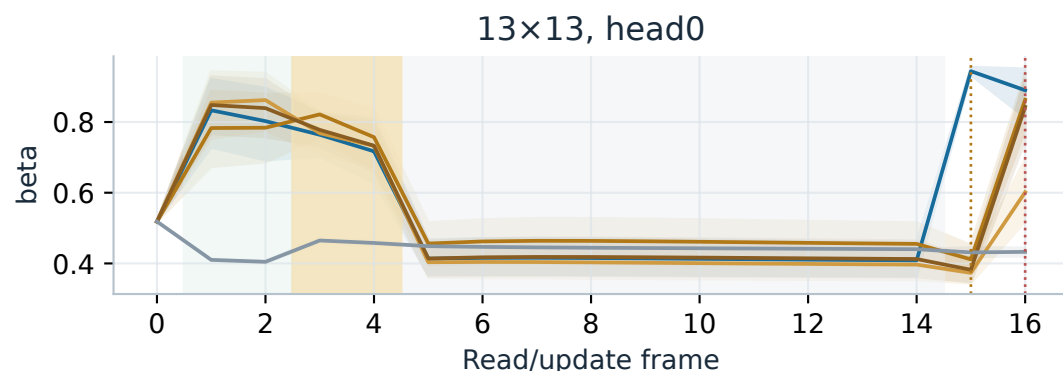
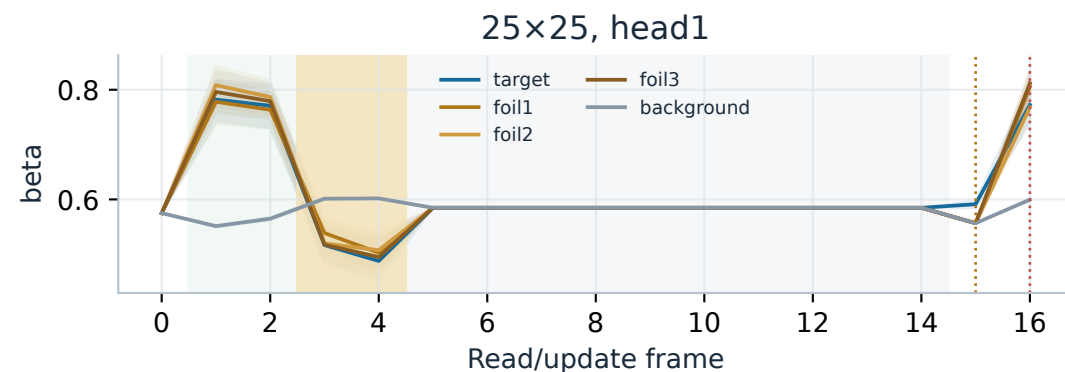
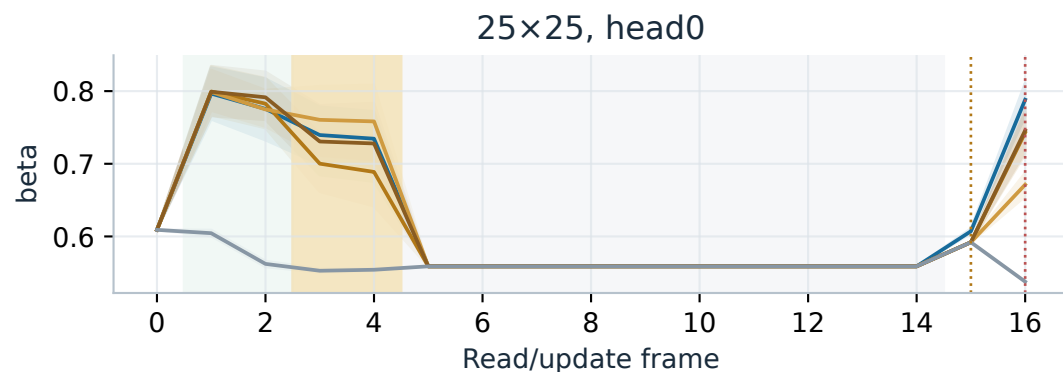
Actual trial0 arrays. Alpha is the mean over 8 retention rows WITHIN each separate head. Signed coefficients can be negative and are neither spatial softmax probabilities nor final-decision attribution. Final-read arrays retain every source frame; fixed-source2 arrays retain every read frame. Stable range per quantity across saved trials/times. Exact weighted-value reconstruction was verified before output convolution.

Downstream 7×7 recurrent readout — distinct measurements



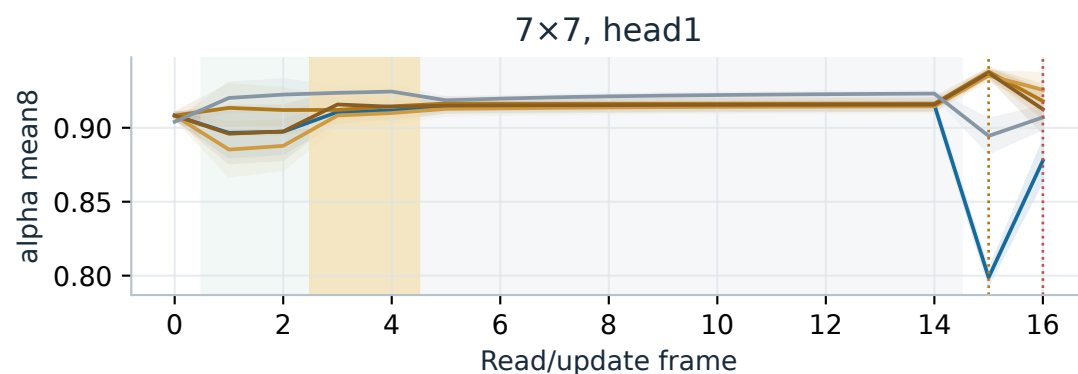
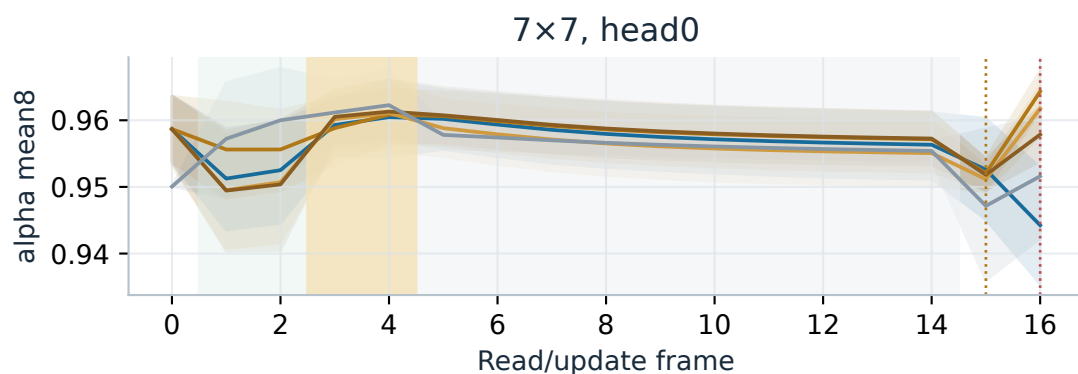
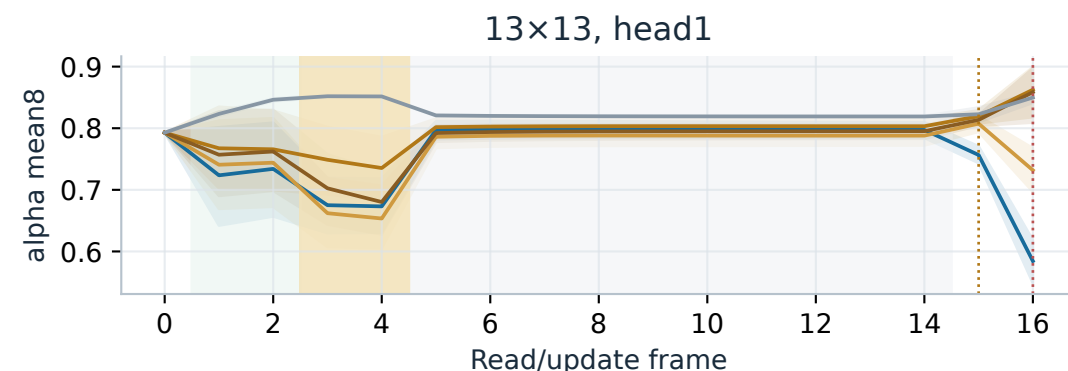
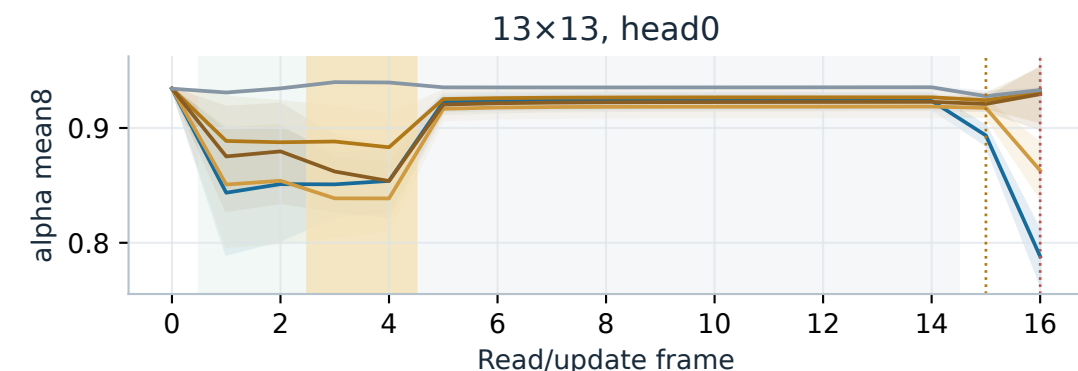
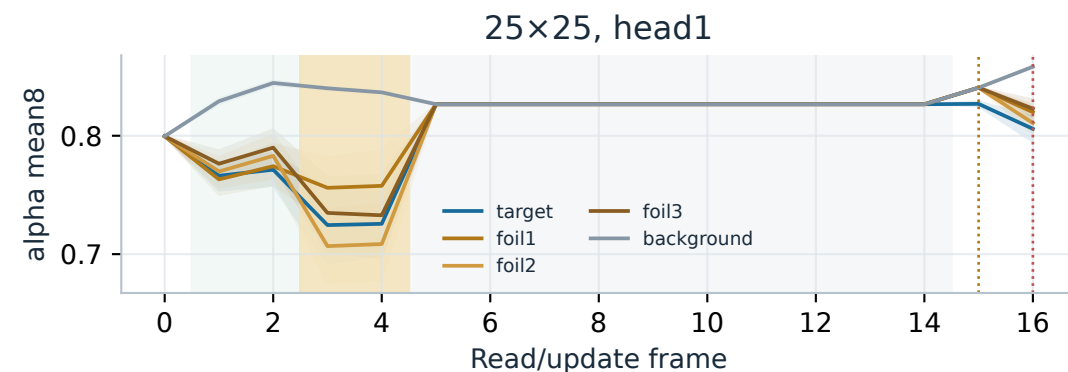
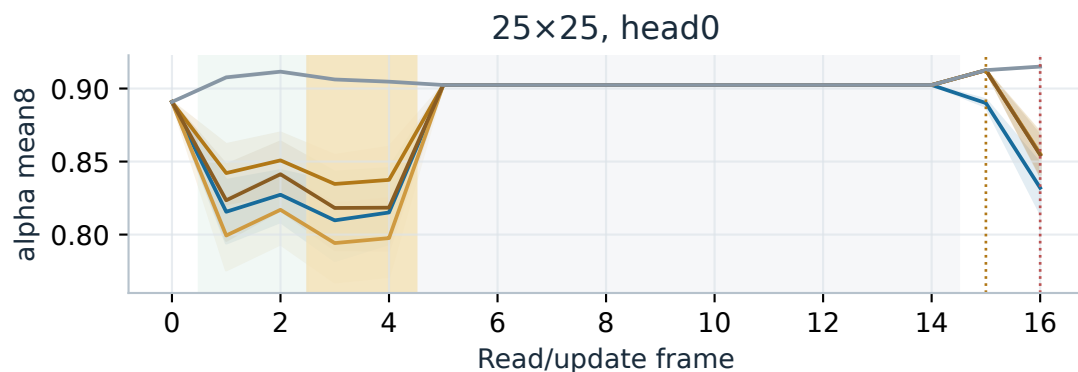
Actual trial0 at frame15. These are ConvGRU write/reset gates and state energy, not KDA gates, implicit coefficients or attention probabilities. Channel averaging is stated explicitly. Full per-timestep arrays retained; no intermediate head is scored as an in-distribution decision.

spatial binding — beta allocation



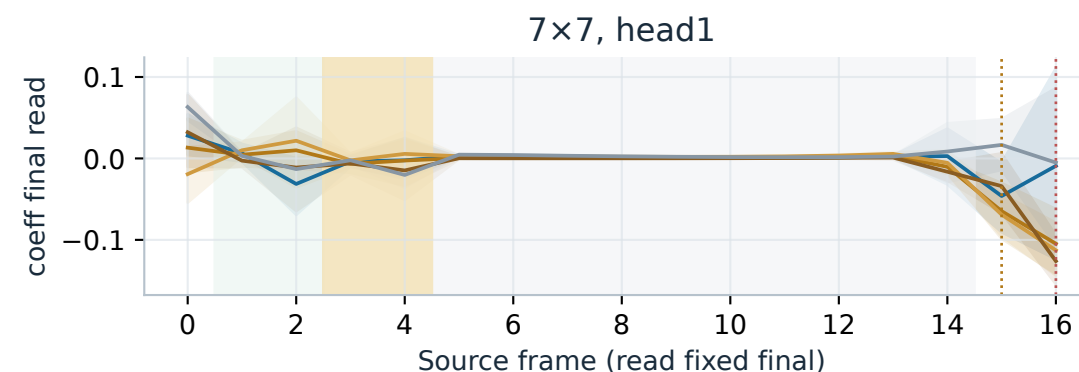
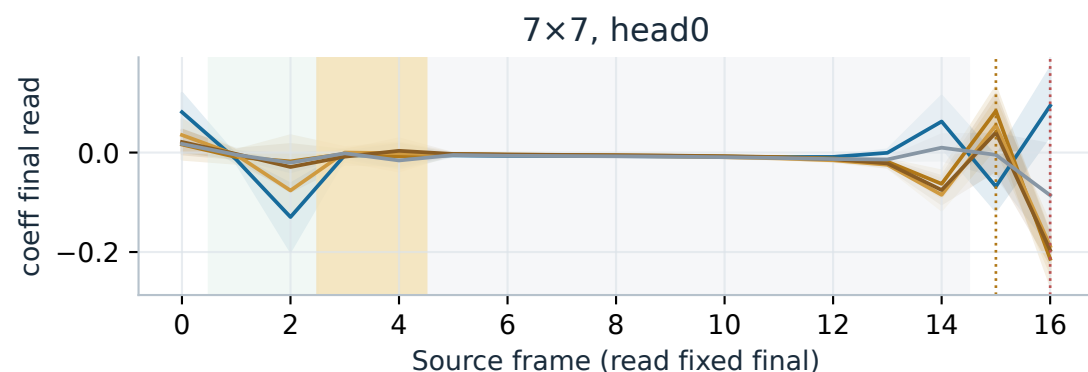
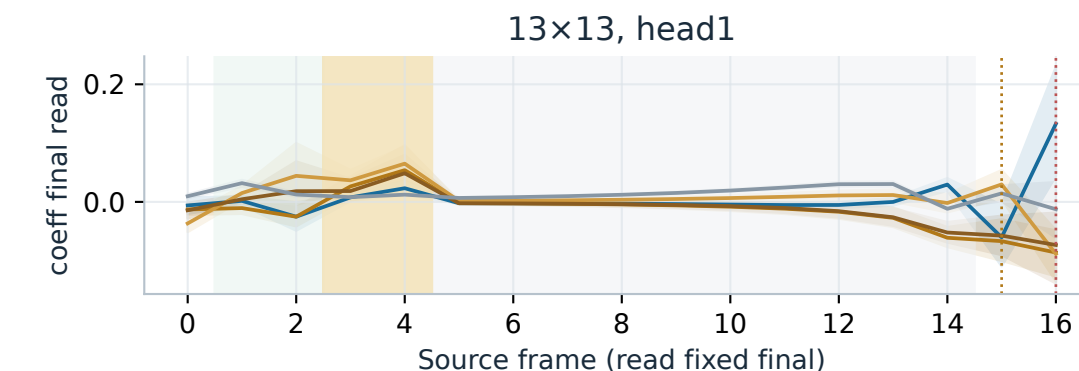
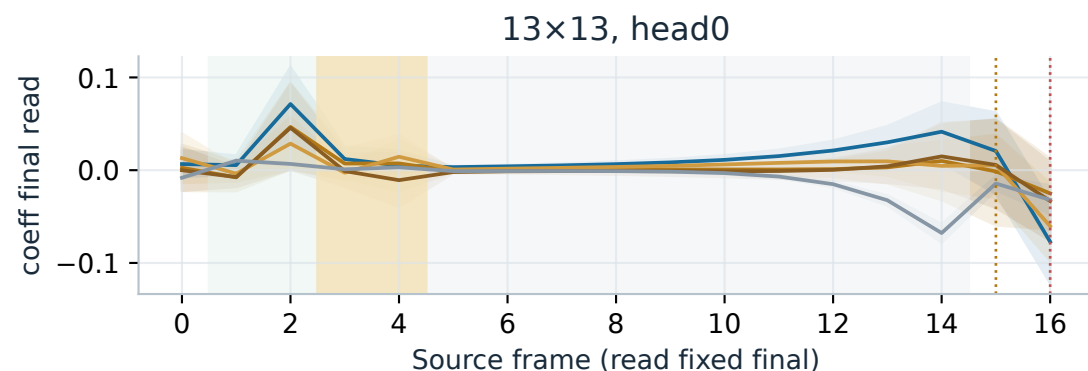
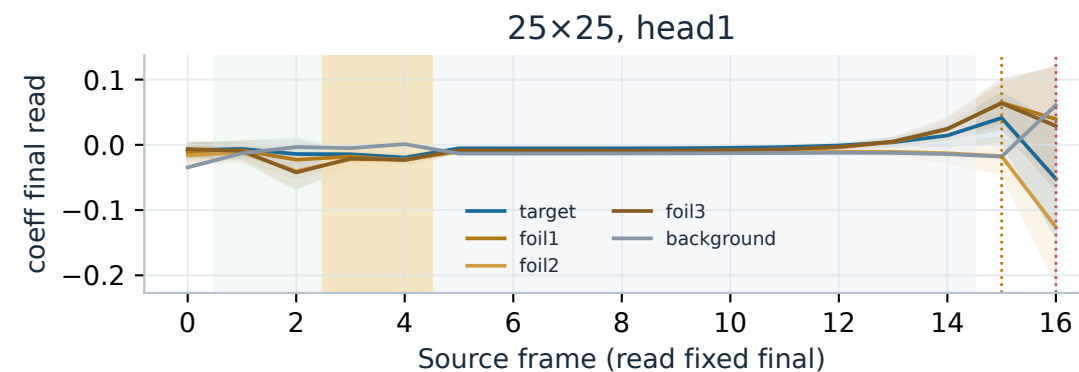
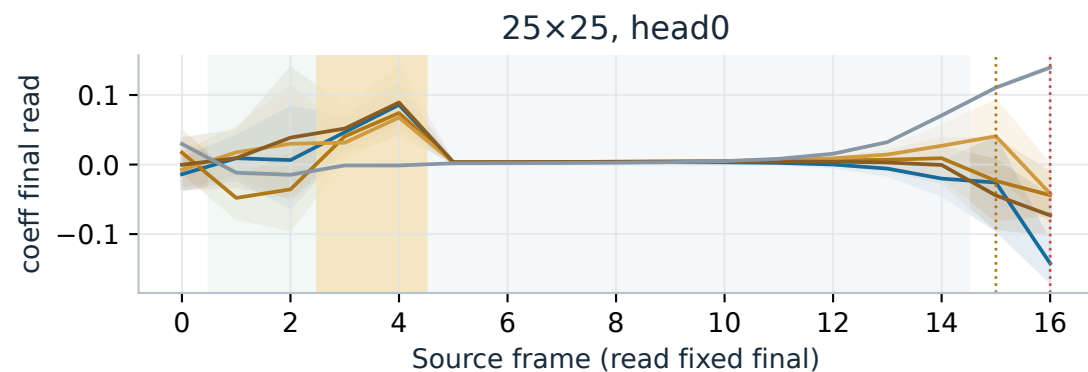
16 independent native D12 trials; each foil separate. Matched translated kernels have equal support/mass at each scale; plotted means are per-mask-mass, not regional totals. Shading is 95% trial-bootstrap uncertainty, never pixels/heads as replicates. Amber phase marks two stack-overlap blanks. For binding, target is only a retrospective eventual-query grouping before the query. Signed coefficient panel preserves sign; separate mean-absolute values are in source CSV.

spatial binding — alpha mean8 allocation



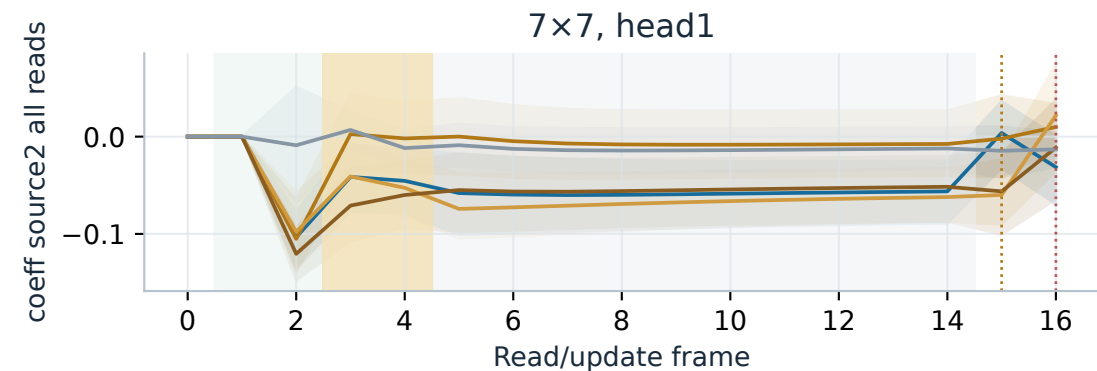
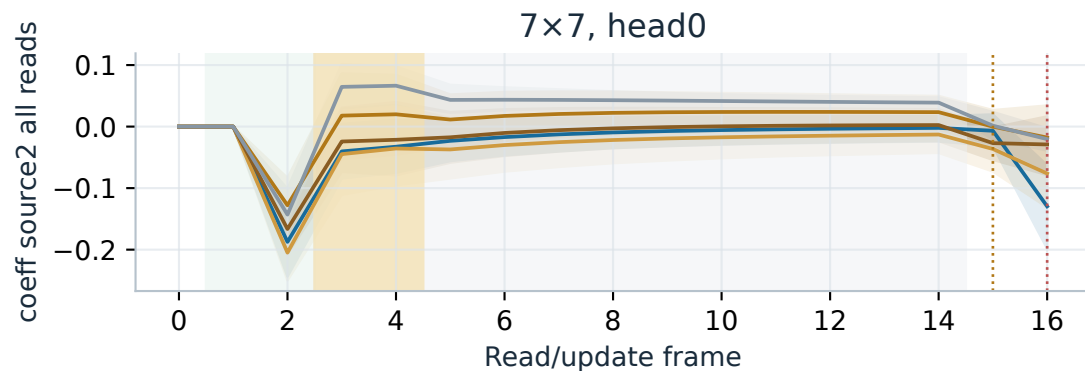
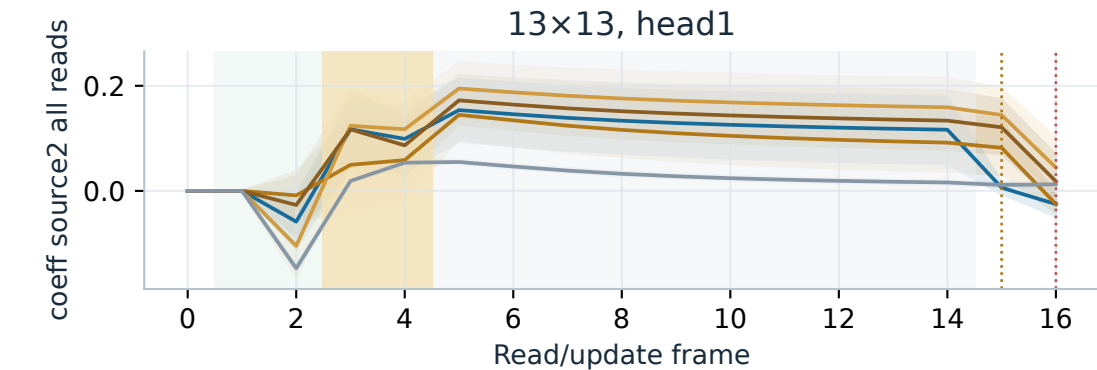
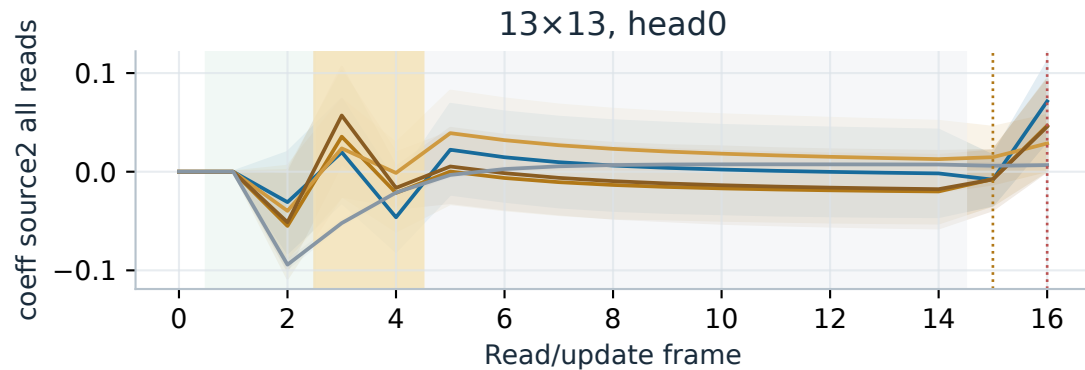
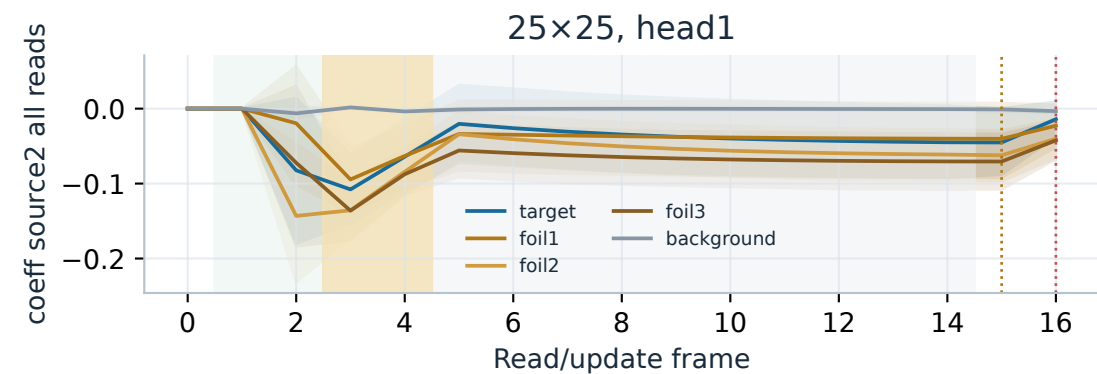
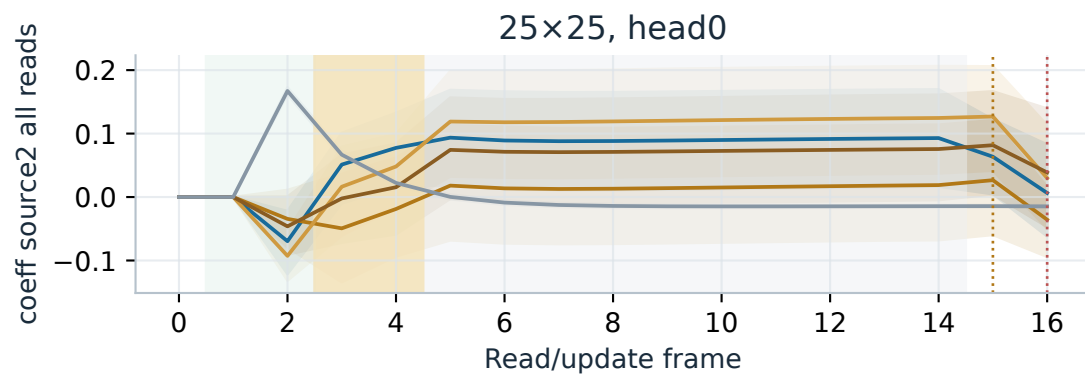
16 independent native D12 trials; each foil separate. Matched translated kernels have equal support/mass at each scale; plotted means are per-mask-mass, not regional totals. Shading is 95% trial-bootstrap uncertainty, never pixels/heads as replicates. Amber phase marks two stack-overlap blanks. For binding, target is only a retrospective eventual-query grouping before the query. Signed coefficient panel preserves sign; separate mean-absolute values are in source CSV.

spatial binding — coeff final read allocation



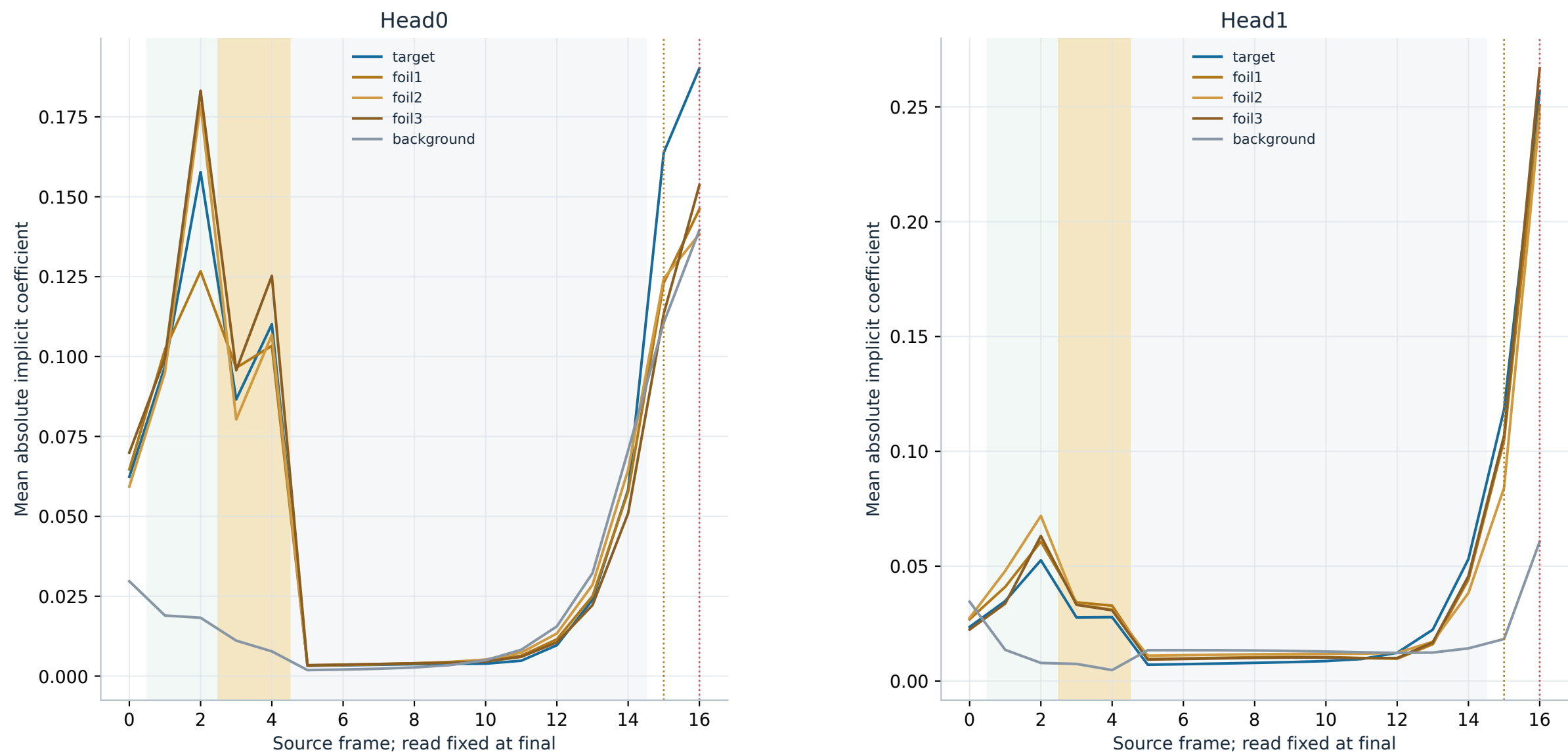
16 independent native D12 trials; each foil separate. Matched translated kernels have equal support/mass at each scale; plotted means are per-mask-mass, not regional totals. Shading is 95% trial-bootstrap uncertainty, never pixels/heads as replicates. Amber phase marks two stack-overlap blanks. For binding, target is only a retrospective eventual-query grouping before the query. Signed coefficient panel preserves sign; separate mean-absolute values are in source CSV.

spatial binding — coeff source2 all reads allocation



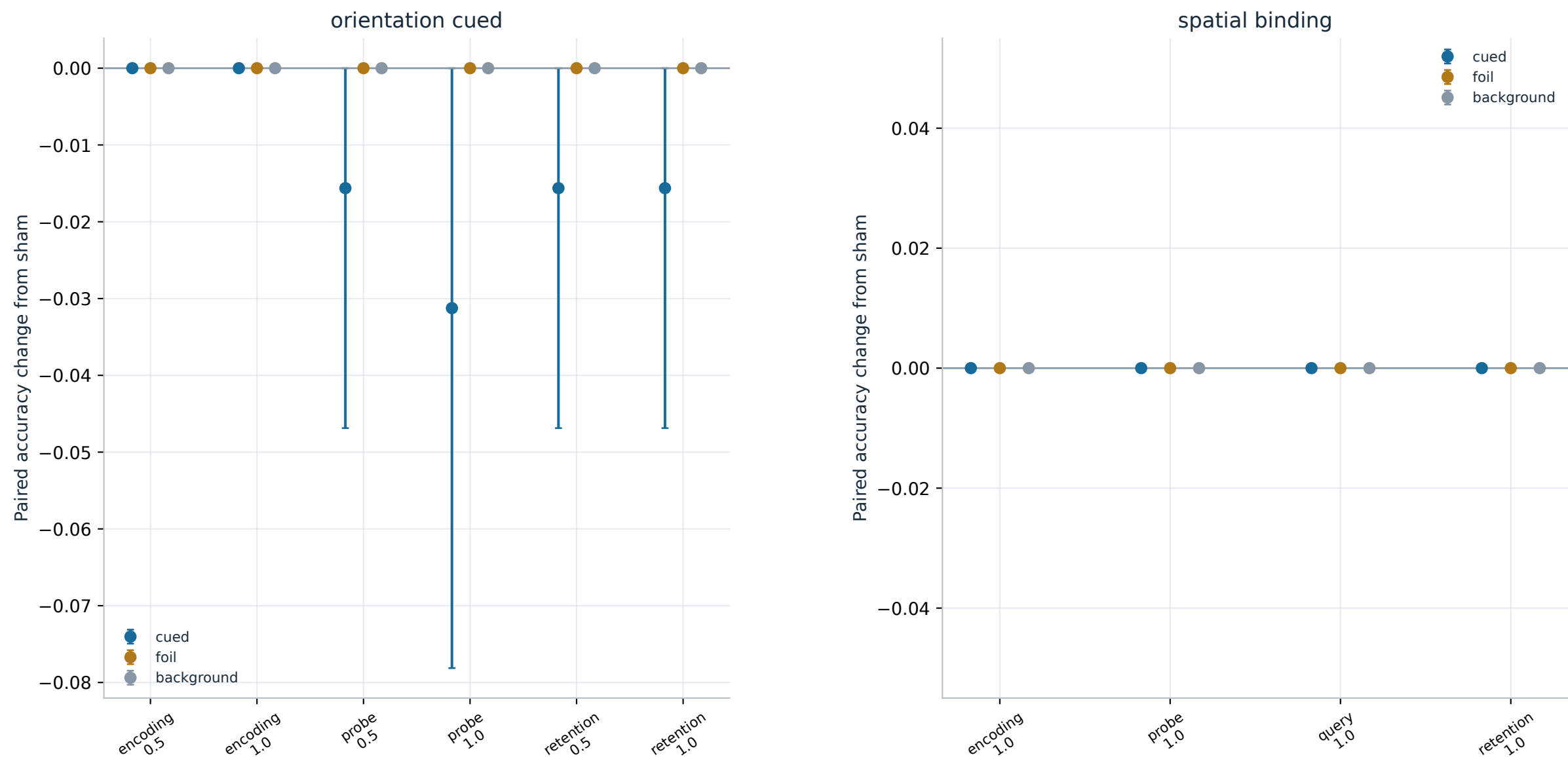
16 independent native D12 trials; each foil separate. Matched translated kernels have equal support/mass at each scale; plotted means are per-mask-mass, not regional totals. Shading is 95% trial-bootstrap uncertainty, never pixels/heads as replicates. Amber phase marks two stack-overlap blanks. For binding, target is only a retrospective eventual-query grouping before the query. Signed coefficient panel preserves sign; separate mean-absolute values are in source CSV.

Absolute coefficient magnitude is not signed allocation



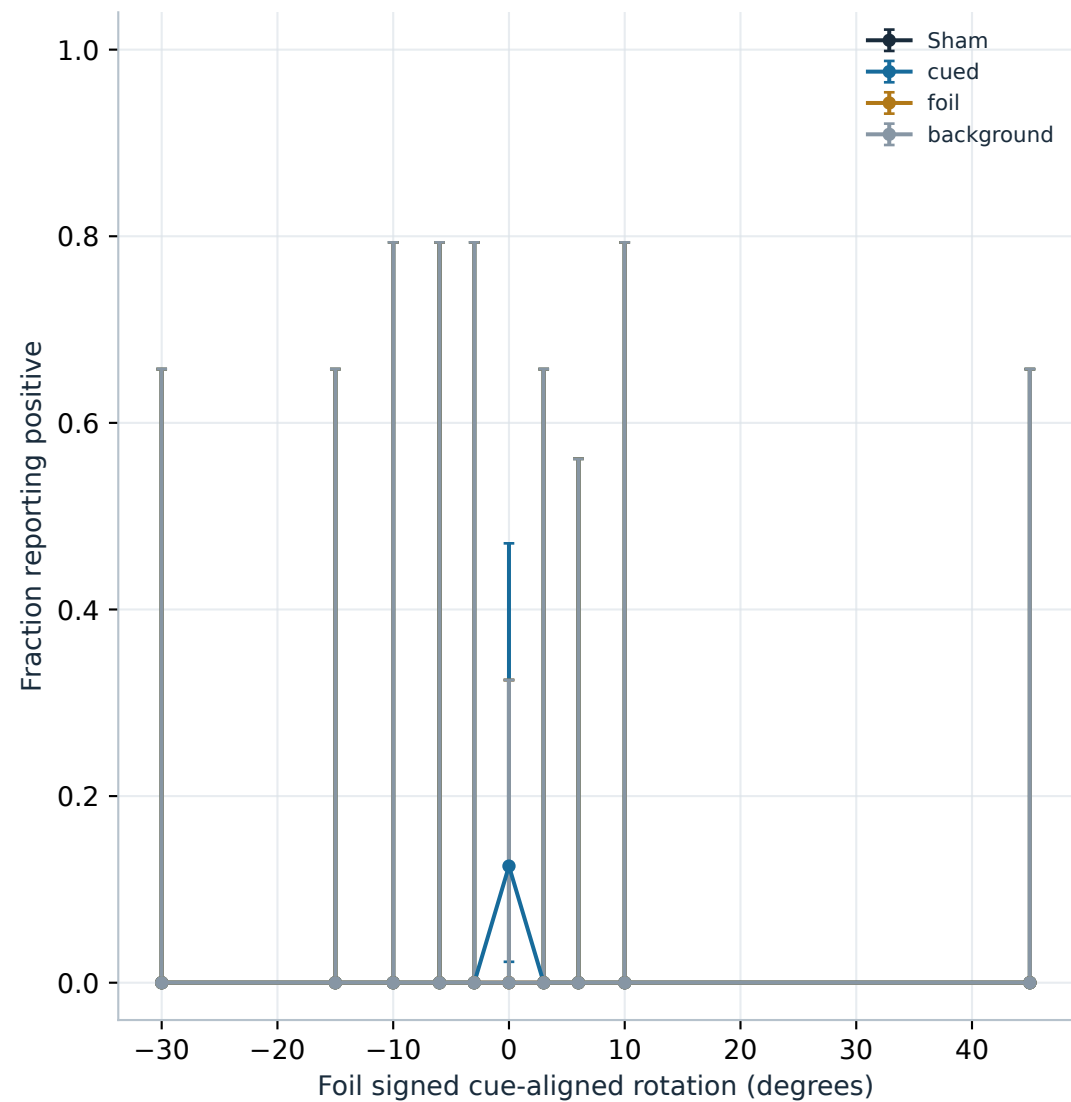
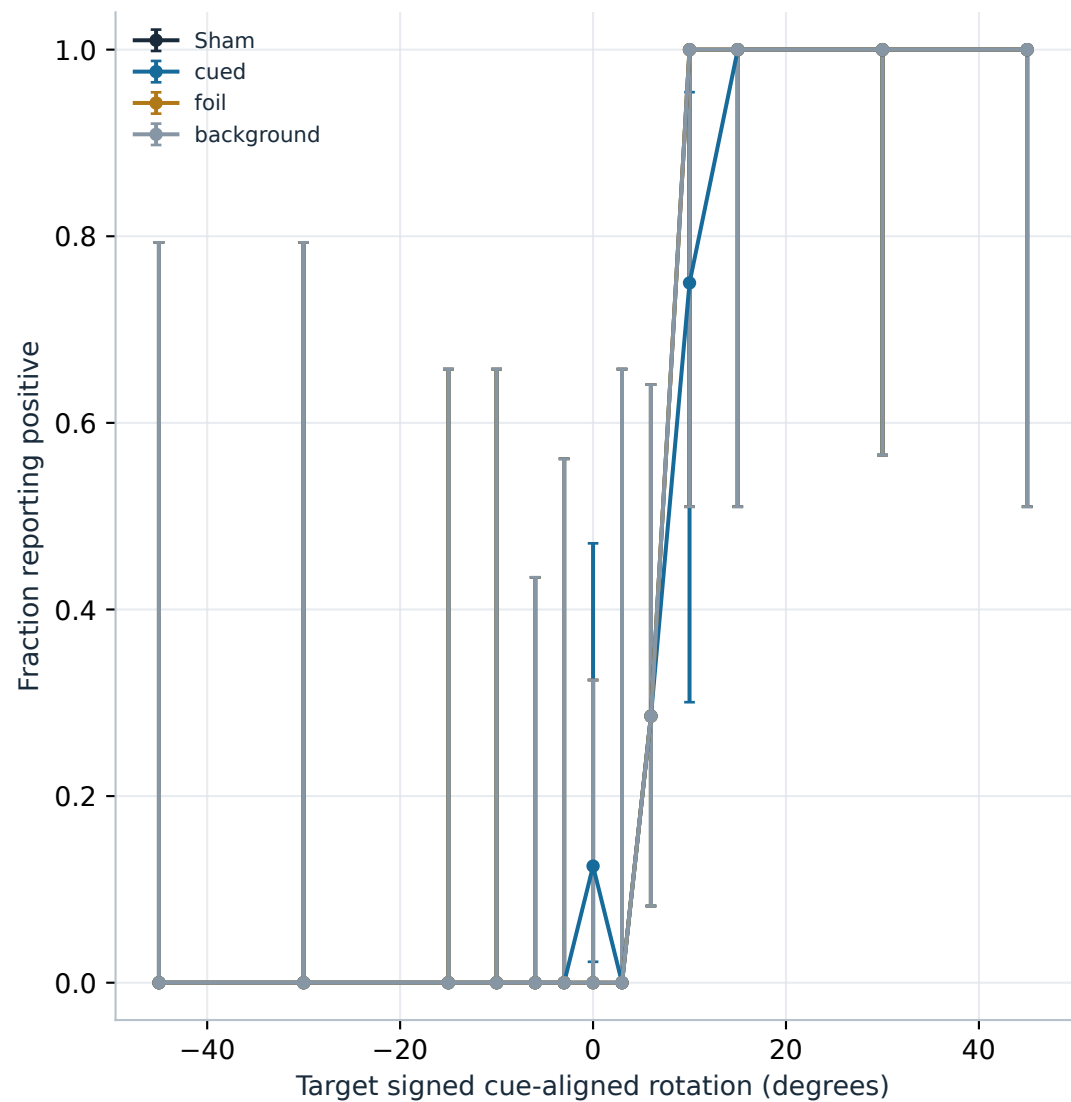
Mean of absolute local coefficients (not absolute of the signed mean). Heads remain separate; source time is not read time. The corresponding signed maps and all-read source2 maps remain in raw arrays and the offline viewer. These quantities describe local memory reads, not patch-to-patch attention or behavioral importance.

Local suppression — site, timing and dose



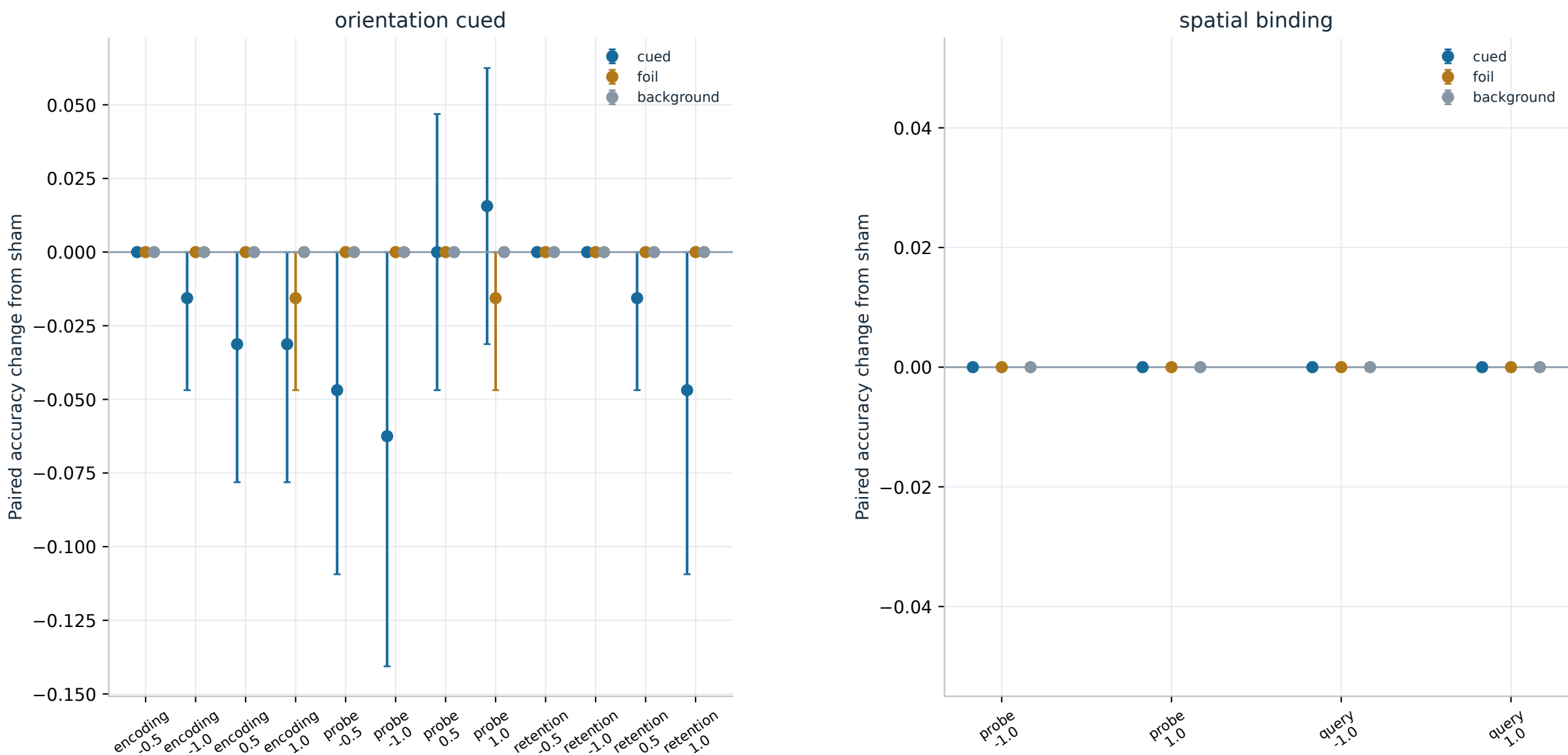
64 paired independent base trials per condition, a prespecified orientation magnitude mixture at D12. One update per pulse. Masks are matched in support/mass, not necessarily removed activation energy; achieved RMS is saved. Error bars 95% paired trial bootstrap; zero-width bars do not prove zero future effect (finite flip bounds in CSV). Binding encoding/retention cued means eventual target, not available cue. Underlying finest-scale KDA state is untouched; downstream recurrence can carry effects.

Suppression psychometrics



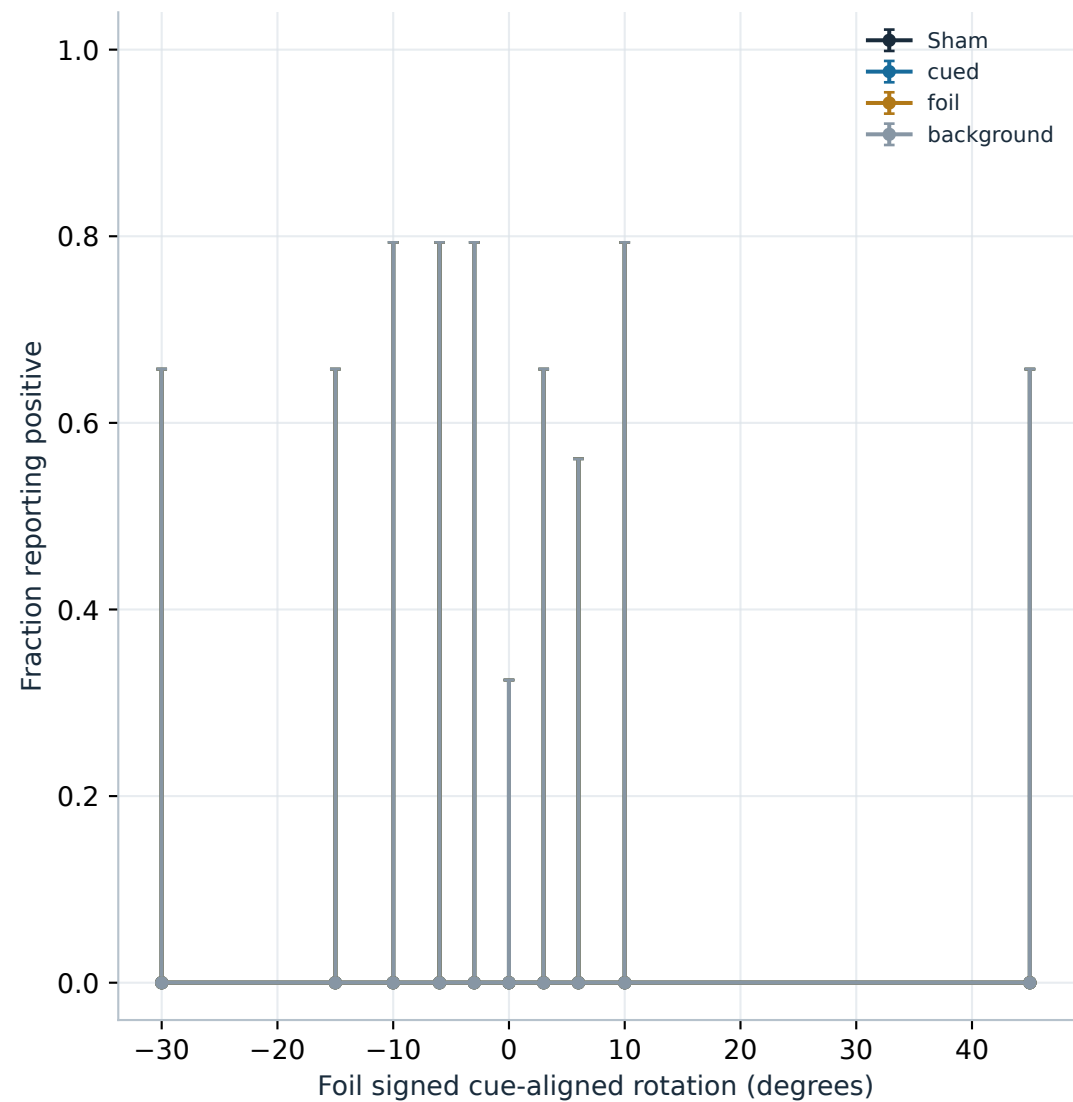
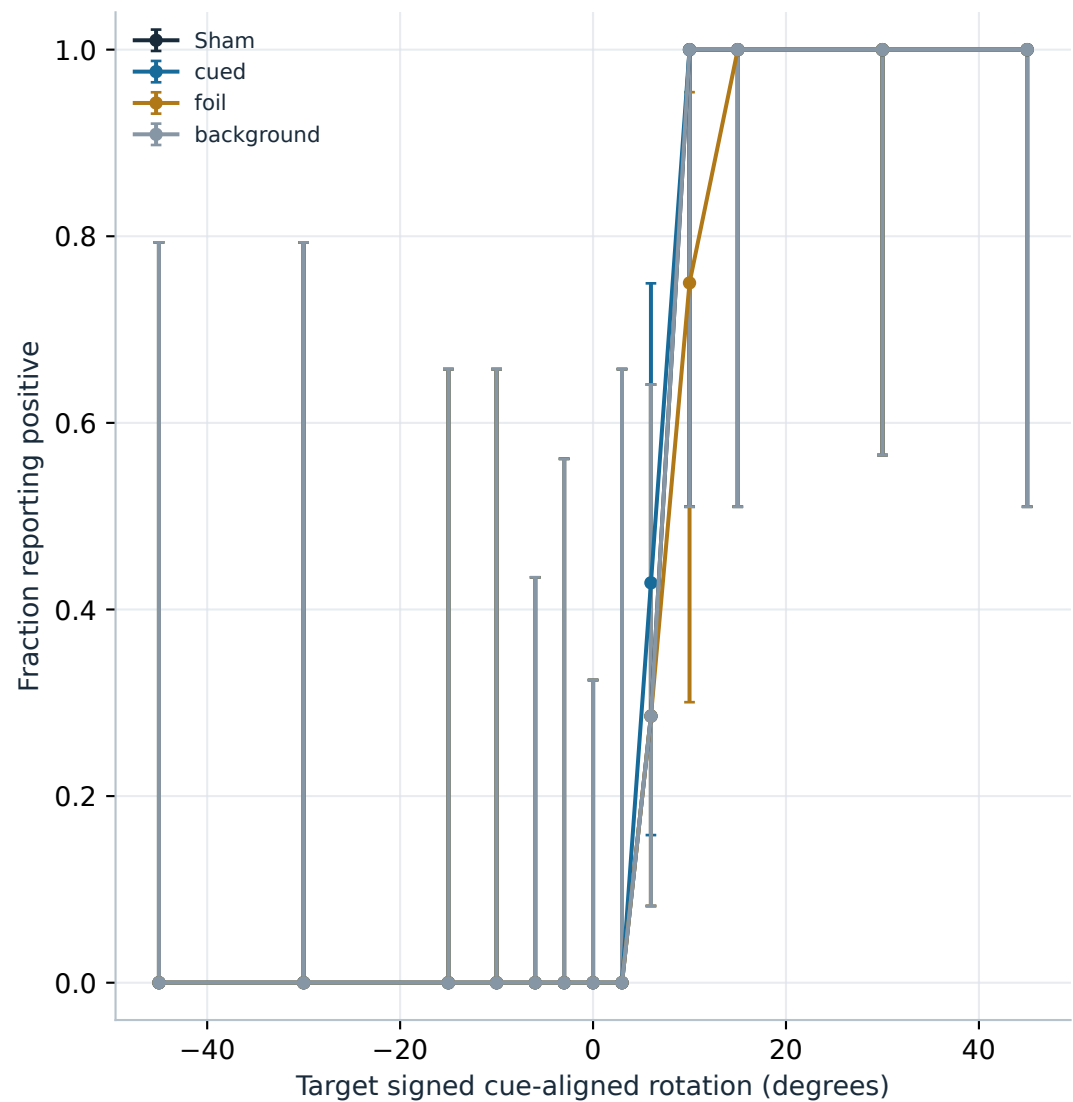
Probe-update dose1 versus the identical sham trials. Left target evidence; right designated foil while target unchanged. Empirical fractions with Wilson95% intervals. Small denominators per magnitude are explicit in trial records:64 TOTAL per condition, not64 per point. No threshold fit or universal benefit is inferred from sparse points. Feature direction is absolute $\sin(2\theta)$, not a learned rotation/decision direction.

Calibrated feature pulses — site, timing and dose



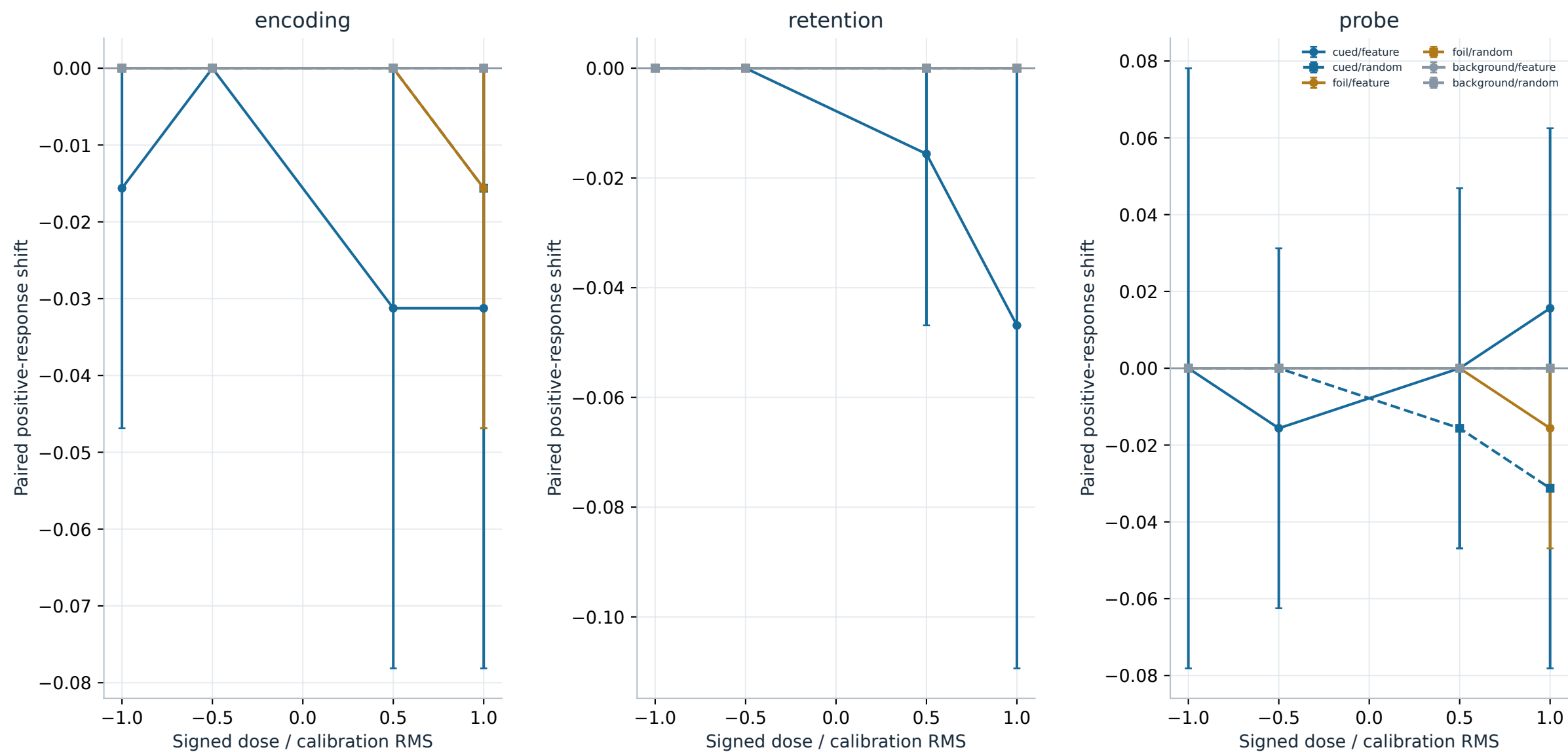
64 paired independent base trials per condition, a prespecified orientation magnitude mixture at D12. One update per pulse. Masks are matched in support/mass, not necessarily removed activation energy; achieved RMS is saved. Error bars 95% paired trial bootstrap; zero-width bars do not prove zero future effect (finite flip bounds in CSV). Binding encoding/retention cued means eventual target, not available cue. Underlying finest-scale KDA state is untouched; downstream recurrence can carry effects.

Positive feature stimulation psychometrics



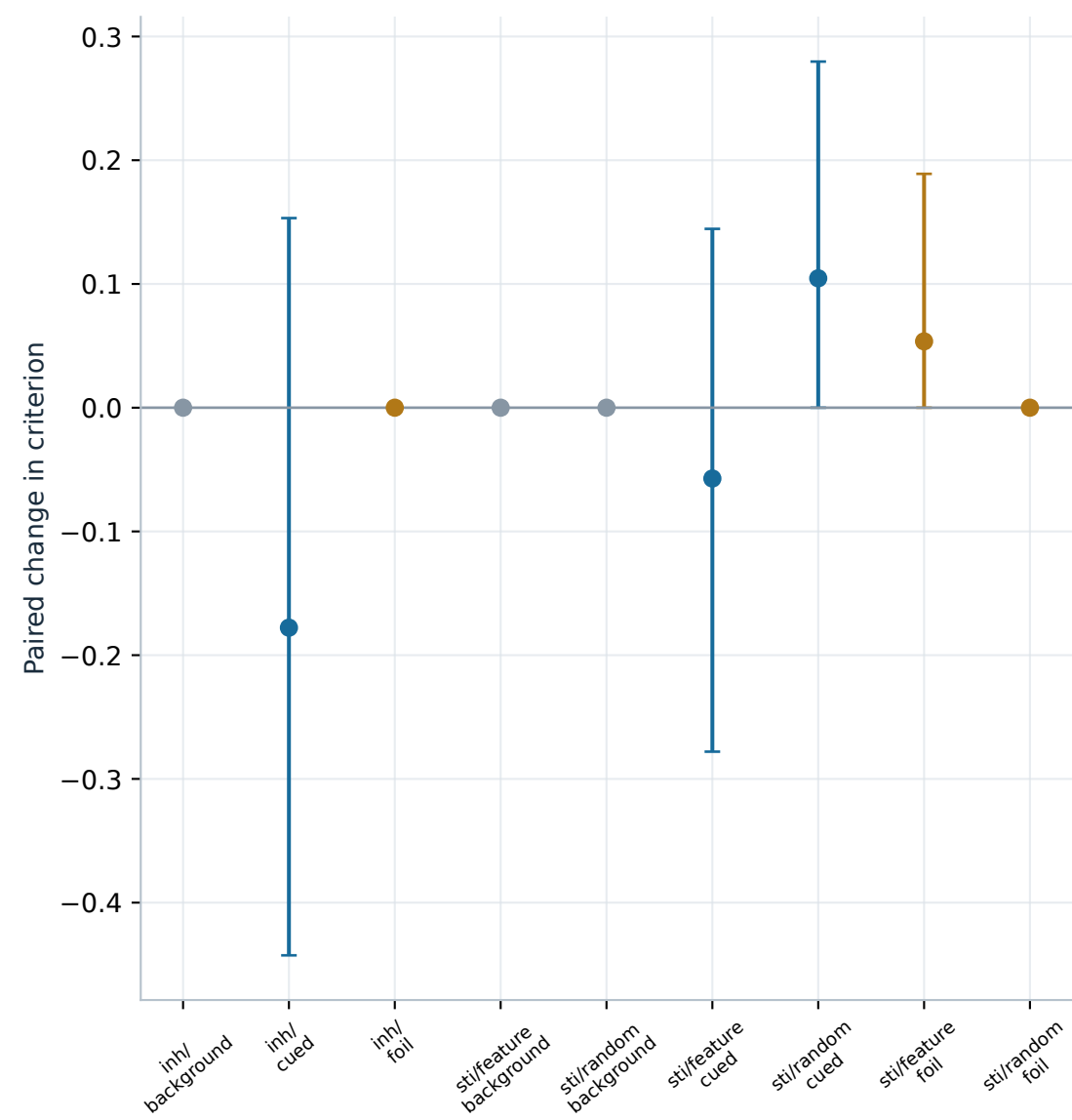
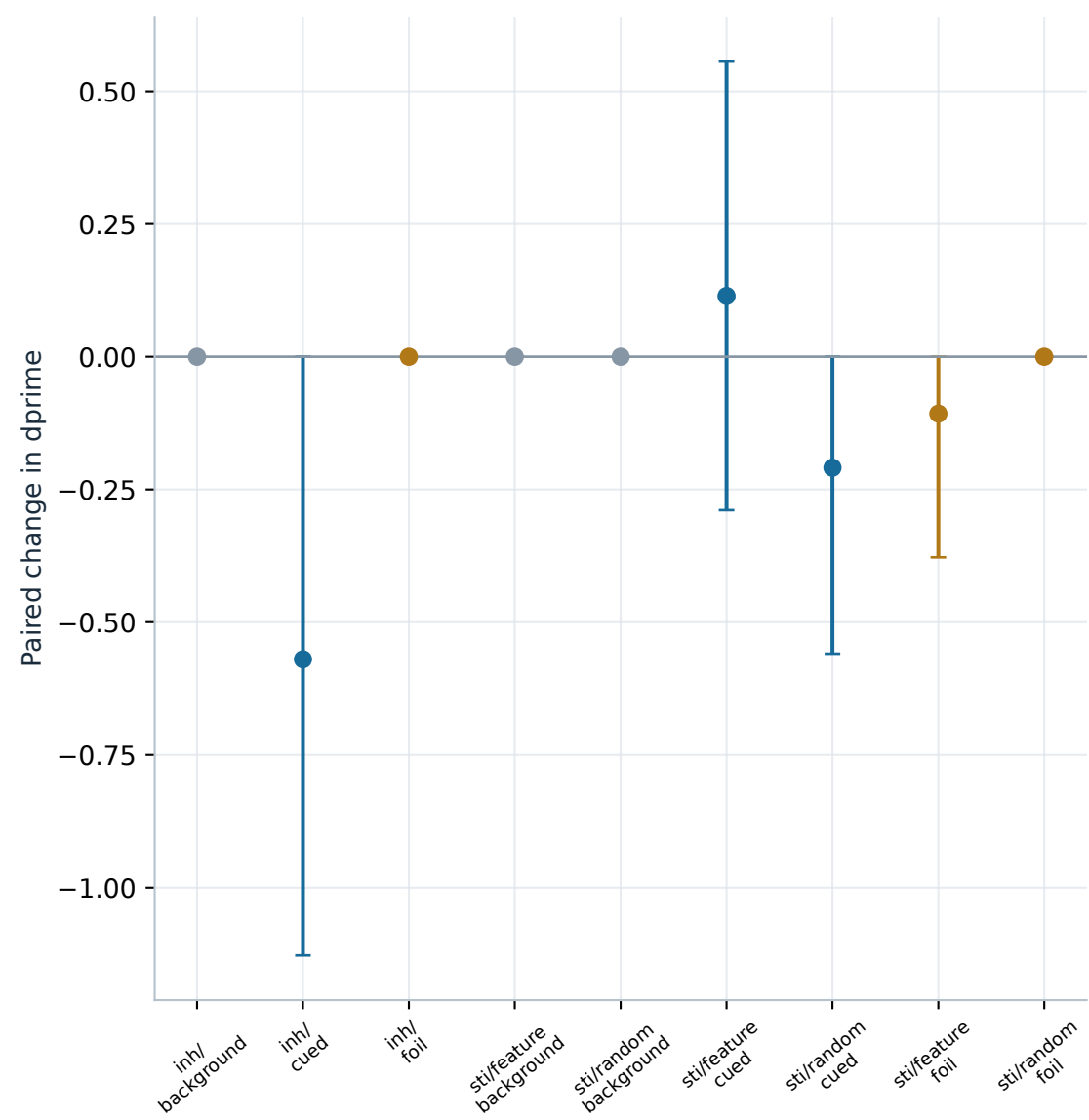
Probe-update dose1 versus the identical sham trials. Left target evidence; right designated foil while target unchanged. Empirical fractions with Wilson95% intervals. Small denominators per magnitude are explicit in trial records:64 TOTAL per condition, not64 per point. No threshold fit or universal benefit is inferred from sparse points. Feature direction is absolute $\sin(2\theta)$, not a learned rotation/decision direction.

Opposite pulses and matched random-direction controls



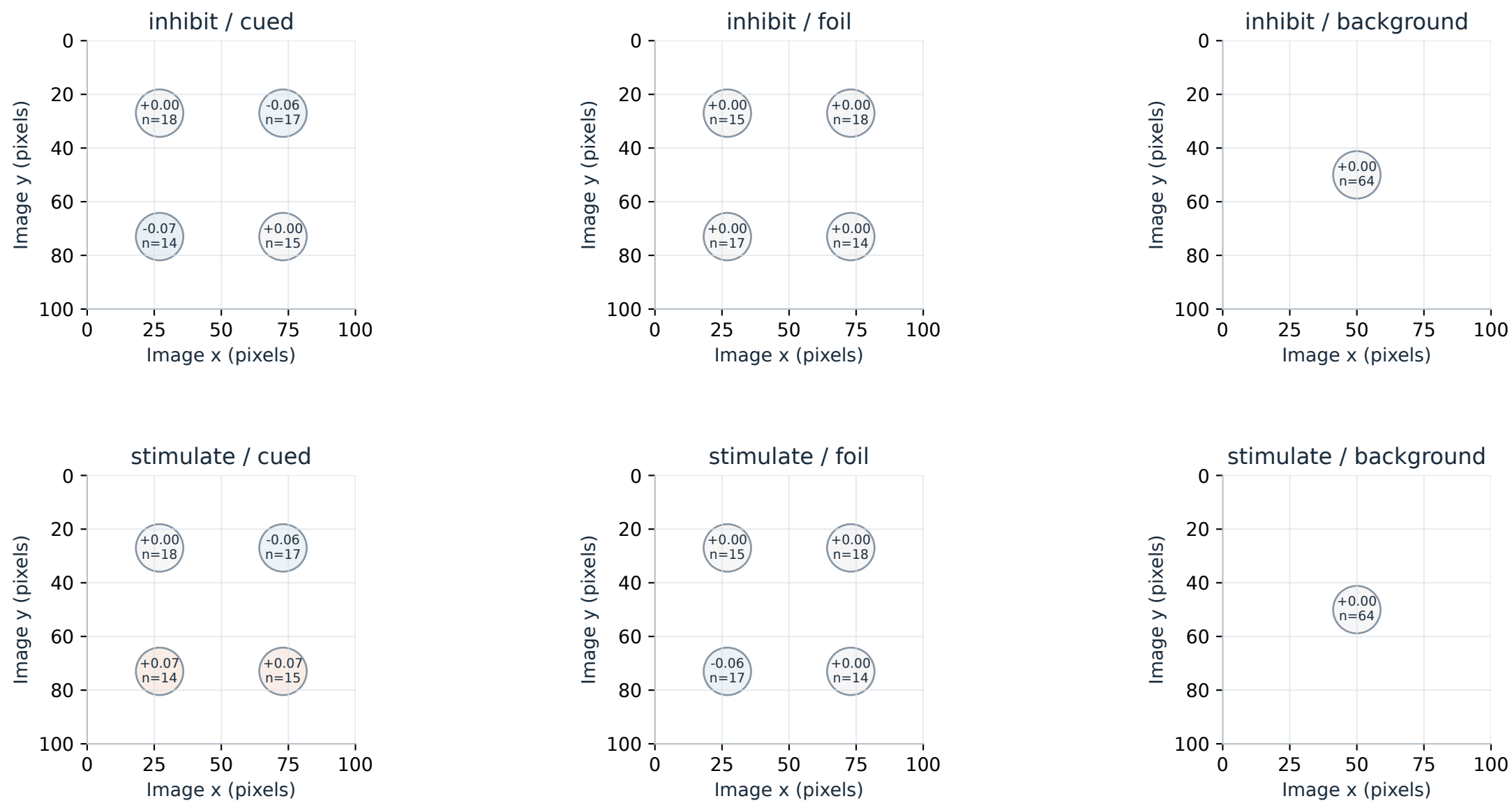
Solid circles: independently calibrated absolute-orientation direction. Dashed squares: orthogonal random direction with identical channel RMS. Positive/negative doses and three matched sites are measured, not inferred. 95% paired bootstrap. A null mean shift can mask opposing sign/location effects; a shift is not necessarily a sensitivity improvement. Physical current, neural cell type and milliseconds are not specified by model units.

Sensitivity and report criterion must be distinguished



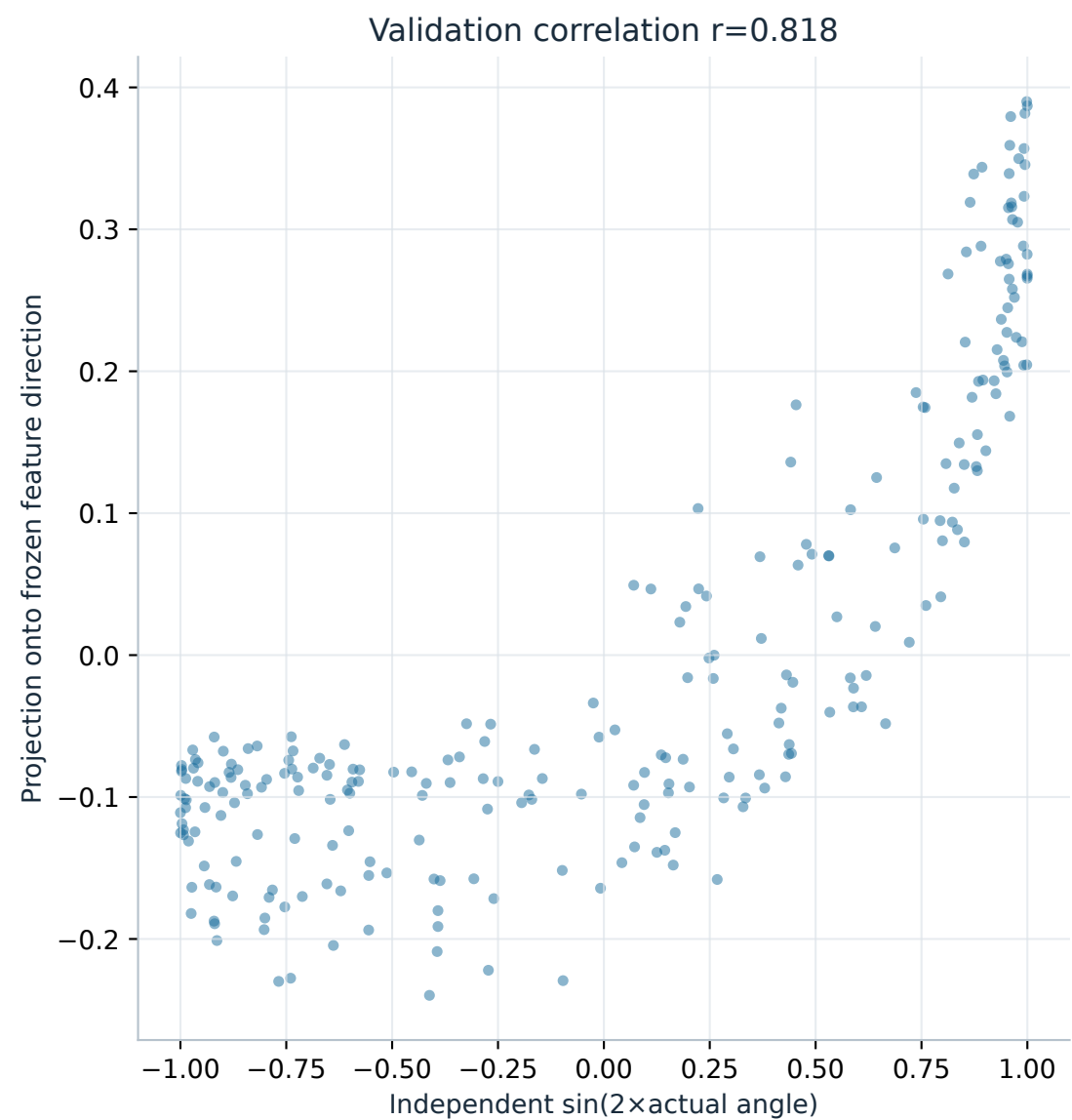
Signal: aligned target rotation. Noise: unchanged or opposite target, pooled explicitly here; separate subtype response rates saved. $d' = Z(\text{hit}) - Z(\text{false alarm})$; criterion = $-[Z(\text{hit}) + Z(\text{false alarm})]/2$. All rates use $\log\text{linear}(k+.5)/(n+1)$ correction, including ceiling cells. 95% paired trial bootstrap; one checkpoint, exploratory multiple contrasts, no familywise significance claim.

Spatial effects: only locations actually perturbed



Top: finest KDA emitted-field suppression¹. Bottom: positive feature stimulation¹RMS. Probe update, orientation D12. Each circle is a tested mask center approximately mapped to image pixels; no interpolation over unmeasured locations. Labels are paired accuracy differences and actual denominators. Smaller physical-location strata are descriptive; relative-site uncertainty is in the paired table.

What the feature pulse does—and does not—identify

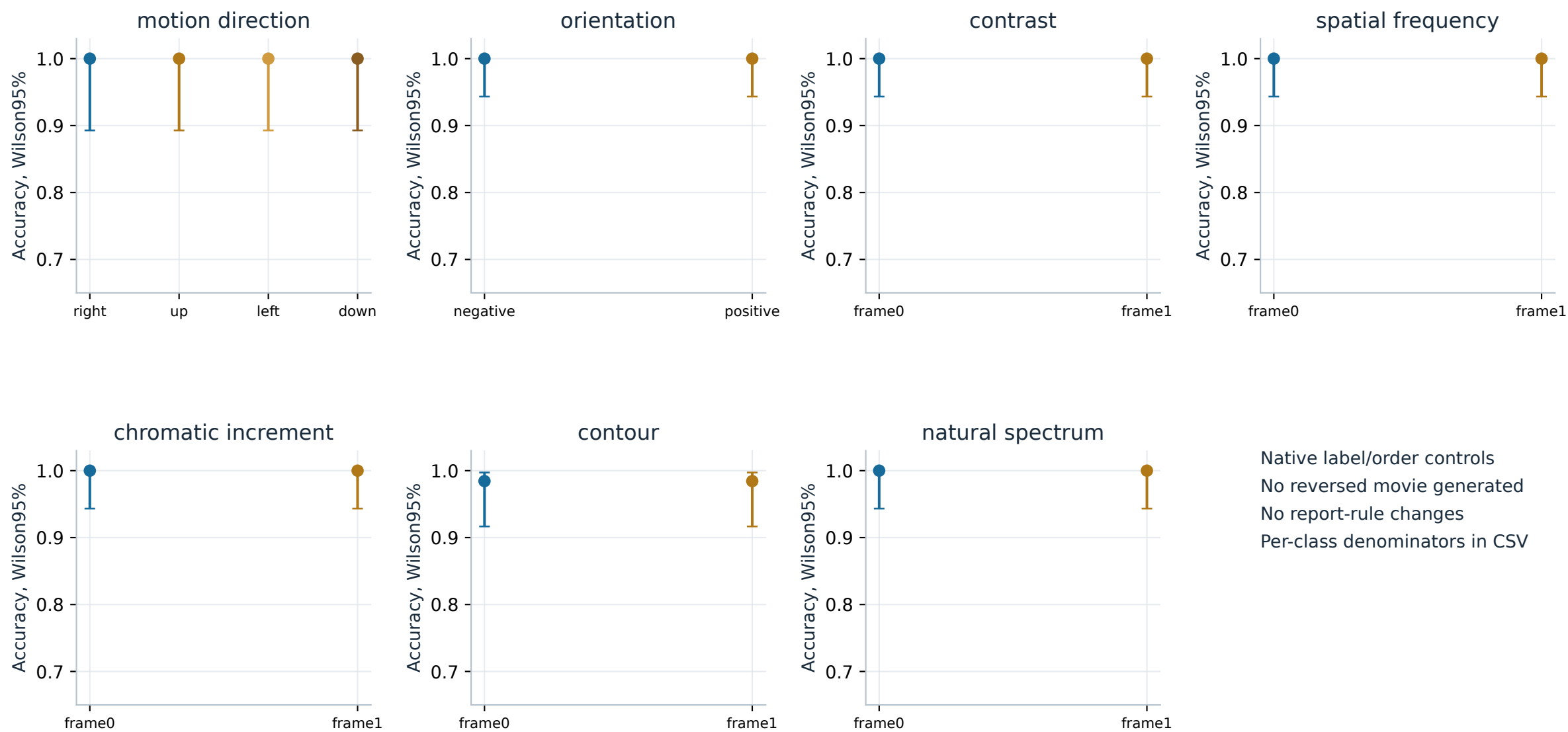


Calibration and causal limits

- 64 fit trials; 64 independent validation trials
- Activation RMS: 0.4574
- 32-channel direction has RMS1
- Absolute orientation preference, not change preference
- No main-model gradients, optimizer or training
- Native final output head only
- One checkpoint; no population-level biological claim

OLS encoding localizer uses actual $\cos(2\theta)/\sin(2\theta)$ orientation on independent native sample frames; trial split precedes pooling locations. Cue location/sign and task label are not calibration targets. Absolute sample orientation is independently drawn from the task label/cue rule. Positive validation confirms feature preference, not a signed rotation comparator or universal behavioral enhancement.

Direction and interval-order controls at native difficulty



Native label/order controls
No reversed movie generated
No report-rule changes
Per-class denominators in CSV

Native observational strata, not synthetic reversals. Motion classes are right/up/left/down; orientation is signed change. Other binary tasks report which frame contains the stronger quantity or structured contour. Points show empirical accuracy and 95% Wilson intervals. These sensory tasks have no spatial cue and cannot establish selective attention by themselves.